UCT MECHATRONICS

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SPES BONA

CHAPTER 2: SYSTEM DYNAMICS

DYNAMIC RESPONSES

OF LTI SYSTEMS

DYNAMIC SYSTEMS

Dynamic system = system that can be modelled using differential equations in time.

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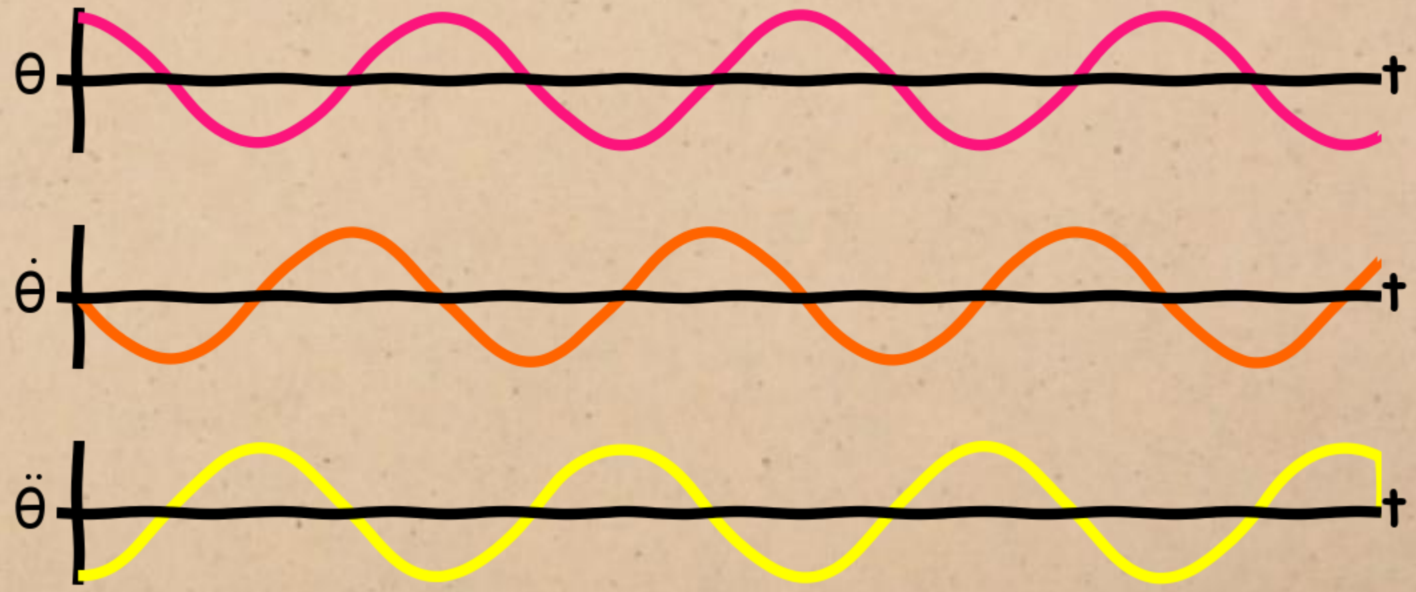
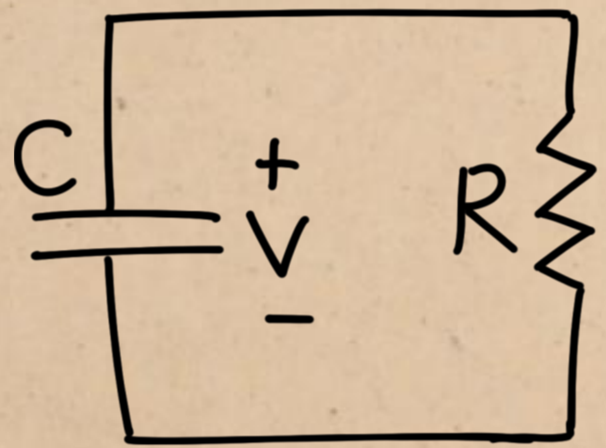


Example 1: description of motion using position, velocity and acceleration.

DYNAMIC SYSTEMS

Dynamic system = system that can be modelled using differential equations in time.



$$\frac{1}{R} V(t) = -C\dot{V}(t)$$

Example 2: resistor-capacitor circuit

DYNAMIC SYSTEMS

LTI **Dynamic system** = system that can be modelled using linear, constant-coefficient differential equations in time. (LCCDEs)

Example: spring-mass-damper

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

(all coefficients are constant)

DYNAMIC SYSTEMS

LTI **Dynamic system** = system that can be modelled using linear, constant-coefficient differential equations in time. (LCCDEs)

counter-example: pendulum

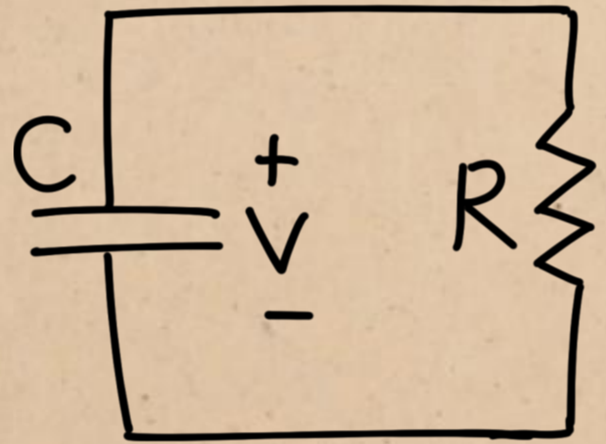
$$\left(I+\frac{1}{4}ml^2\right)\ddot{\theta}(t) = \frac{1}{2}gml\sin(\theta(t))$$

(some coefficients are functions of the variables)

DYNAMIC SYSTEMS

order of differential equation = highest derivative present

first-order:



$$\frac{1}{R}V(t) = -C\dot{V}(t)$$

first derivative

second-order:

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

second derivative

SOLVING LCCDE'S

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

Laplace transform of derivative:

don't forget the initial conditions!

properties

	Transform Pair/Property
	$ax(t) + bv(t) \iff aX(s) + bV(s)$
	$x(t-a)u(t-a) \iff e^{-as}X(s) \quad a \geq 0$
	$x(at) \iff \frac{1}{a}X(\frac{s}{a}) \quad a > 0$
	$t^n x(t) \iff (-1)^n X^{(n)}(s)$
	$e^{at}x(t) \iff X(s-a)$
	$x'(t) \iff sX(s) - \underline{x(0^-)}$
	$x''(t) \iff s^2X(s) - \underline{sx(0^-) - x'(0^-)}$
	$x^{(n)}(t) \iff s^nX(s) - \underline{s^{n-1}x(0^-) - \cdots - x^{(n-1)}(0^-)}$
Integration	$\int_{0^-}^{t} x(\lambda)d\lambda \iff \frac{1}{s}X(s)$
	$\int_{-\infty}^{t} x(\lambda)d\lambda \iff \frac{1}{s}X(s) + \frac{1}{s}\int_{-\infty}^{0^-} x(\lambda)d\lambda$
Time convolution	$x(t) * v(t) \iff X(s)V(s)$
Frequency convolution	$x(t)v(t) \iff \frac{1}{2\pi j}X(\omega) * V(\omega)$

Initial value: $f(0^+) = \lim_{s\to\infty} sF(s)$

Final value: $f(\infty) = \lim_{s\to 0} sF(s)$ with all poles in left-hand plane

Common Unilateral Laplace Transform Pairs

$x(t) = \frac{1}{2\pi j}\int_{c-j\infty}^{c+j\infty} X(s)e^{st}ds$	$X(s) = \int_{0^-}^{\infty} x(t)e^{-st}dt$
$\delta(t)$	1

SOLVING LCCDE'S

Differential equations in the time domain become algebraic equations in the s domain.

$$m(s^2x(s) - sx(0) - \dot{x}(0)) = -b(sx(s) - x(0)) - kx(s) + F(s)$$

SOLVING LCCDE'S

Differential equations in the time domain become algebraic equations in the s domain.

$$ms^2x(s) - msx(0) - m\dot{x}(0) = -bsx(s) + bx(0) - kx(s) + F(s)$$

SOLVING LCCDE'S

Differential equations in the time domain become algebraic equations in the s domain.

$$ms^2x(s) + bsx(s) + kx(s) = F(s) + bx(0) + msx(0) + m\dot{x}(0)$$

SOLVING LCCDE'S

Differential equations in the time domain become algebraic equations in the s domain.

$$(ms^2 + bs + k)x(s) = F(s) + (b + ms)x(0) + m\dot{x}(0)$$

DYNAMIC RESPONSE

forced response = part of response

related to input

(particular solution)

$$x(s) = \frac{1}{ms^2 + bs + k} F(s) + \frac{(b + ms)x(0) + m\dot{x}(0)}{ms^2 + bs + k}$$

free response = part of response

related to initial conditions

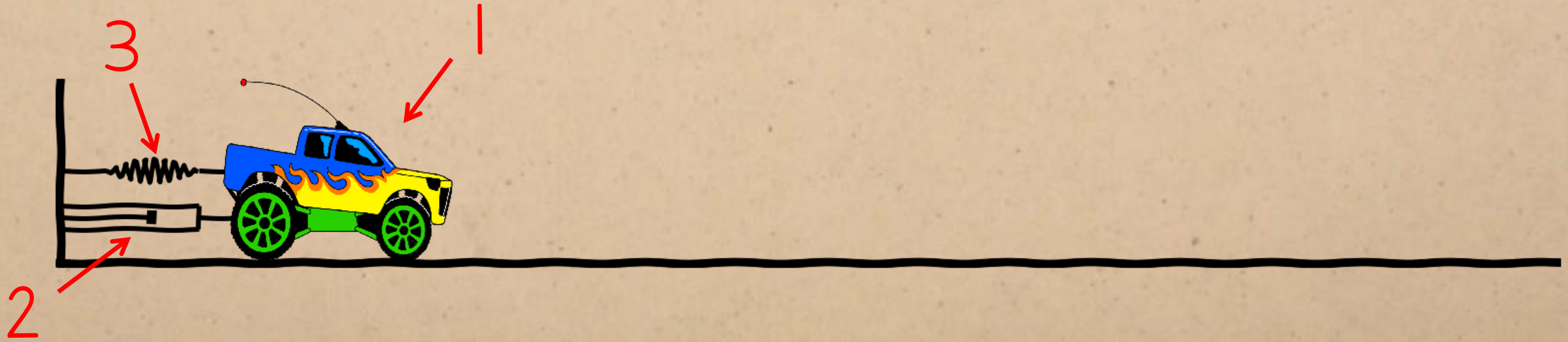
(homogenous solution)

DYNAMIC RESPONSE

Using specific parameters...

Let $m = 1$, $b = 3$ and $k = 2$:

$$x(s) = \frac{1}{s^2 + 3s + 2}F(s) + \frac{(3 + s)x(0) + \dot{x}(0)}{s^2 + 3s + 2}$$



DYNAMIC RESPONSE

characteristic equation = denominator of response set equal to zero

$$x(s) = \frac{1}{s^2 + 3s + 2} F(s) + \frac{(3 + s)x(0) + \dot{x}(0)}{s^2 + 3s + 2}$$

$$s^2 + 3s + 2 = 0$$

DYNAMIC RESPONSE

poles = roots of the characteristic equation

$$x(s) = \frac{1}{(s+1)(s+2)}F(s) + \frac{(3+s)x(0) + \dot{x}(0)}{(s+1)(s+2)}$$

2 poles

$$(s - -1)(s - -2) = 0$$

number of poles = **order** of system

second-order

$$m\ddot{x}(t) = -b\dot{x}(t) - kx(t) + F(t)$$

DYNAMIC RESPONSE

rest = zero initial conditions

$$x(s) = \frac{1}{(s+1)(s+2)}F(s) + \frac{(3+s)0+0}{(s+1)(s+2)}$$

DYNAMIC RESPONSE

transfer function

$$x(s) = \frac{1}{(s+1)(s+2)} F(s)$$

DYNAMIC RESPONSE

To find the time-domain response, we start by splitting up the transfer function using partial fraction expansion...

$$x(s) = \frac{1}{(s+1)(s+2)} F(s)$$

$$\frac{A}{s+1} + \frac{B}{s+2} = \frac{1}{(s+1)(s+2)} \quad \therefore A(s+2) + B(s+1) = 1$$

let $s = -1$: $A(-1+2) + B(-1+1) = 1 \therefore A = 1$

let $s = -2$: $A(-2+2) + B(-2+1) = 1 \therefore -B = 1 \therefore B = -1$

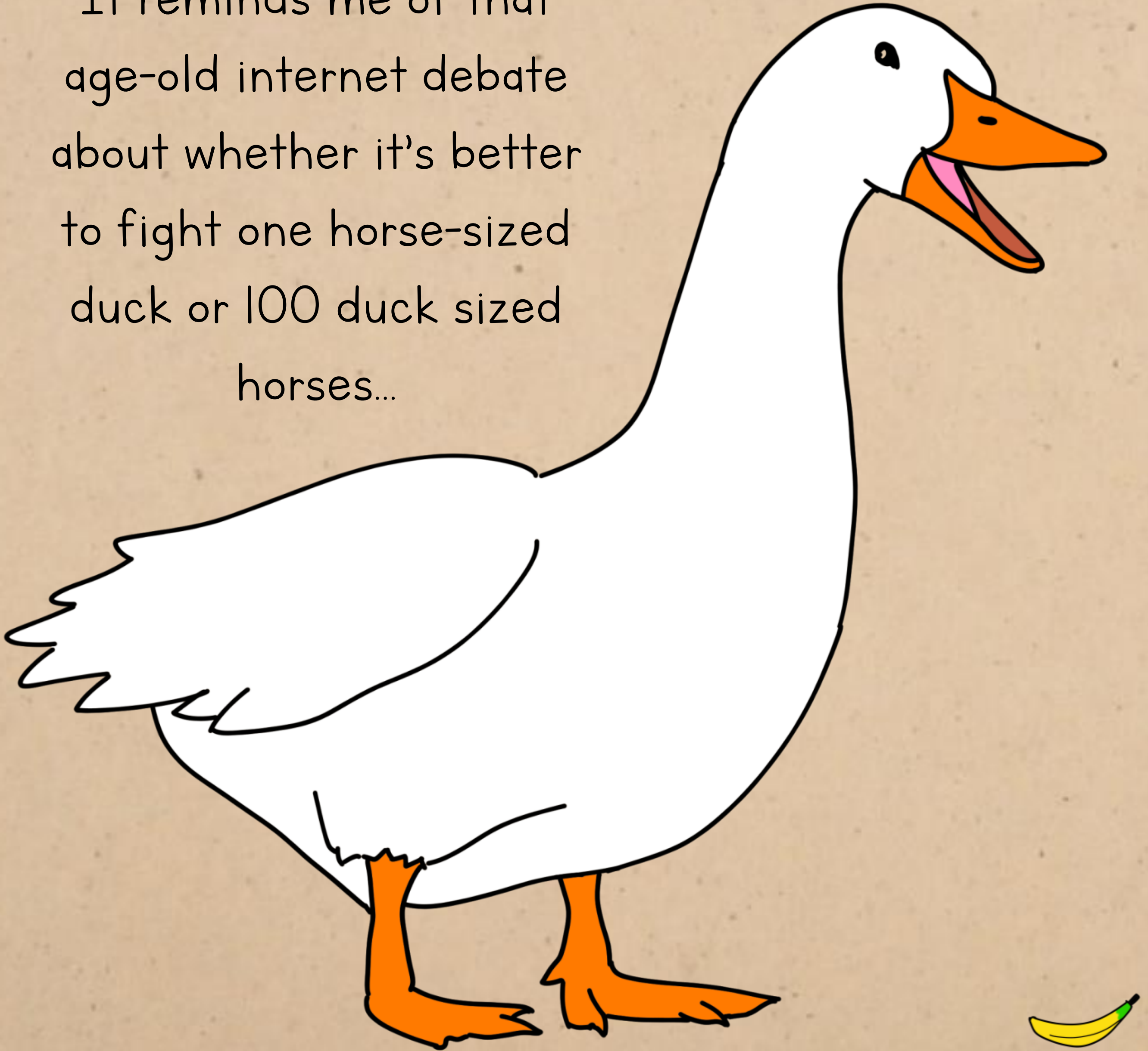
$$\therefore \frac{1}{s+1} - \frac{1}{s+2}$$

DYNAMIC RESPONSE

$$x(s) = \frac{1}{s+1}F(s) - \frac{1}{s+2}F(s)$$

This is a very important and powerful step, because now, instead of thinking of the system response as a complicated, monolithic thing, we're working with a sum of simple, first-order parts.

It reminds me of that age-old internet debate about whether it's better to fight one horse-sized duck or 100 duck sized horses...



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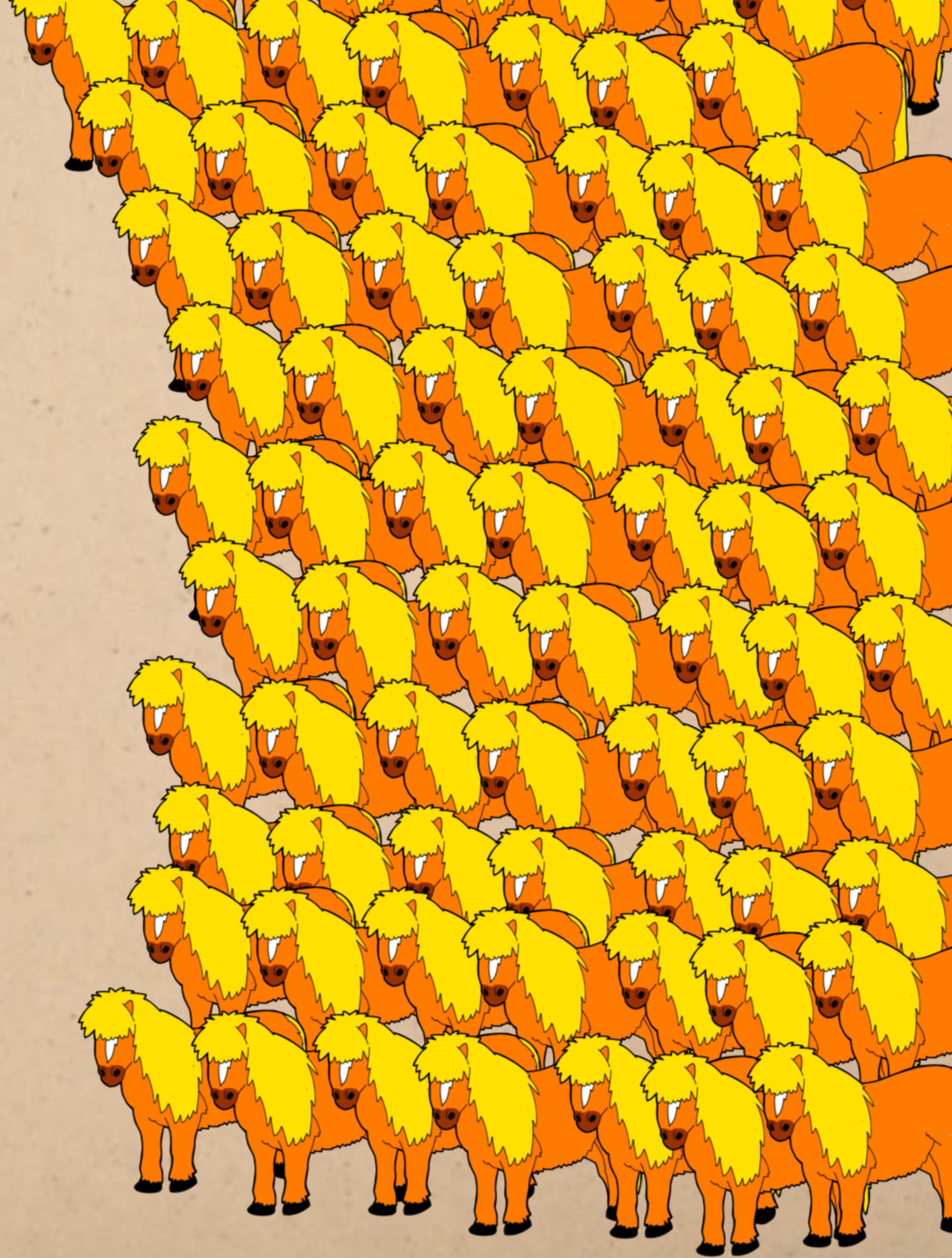
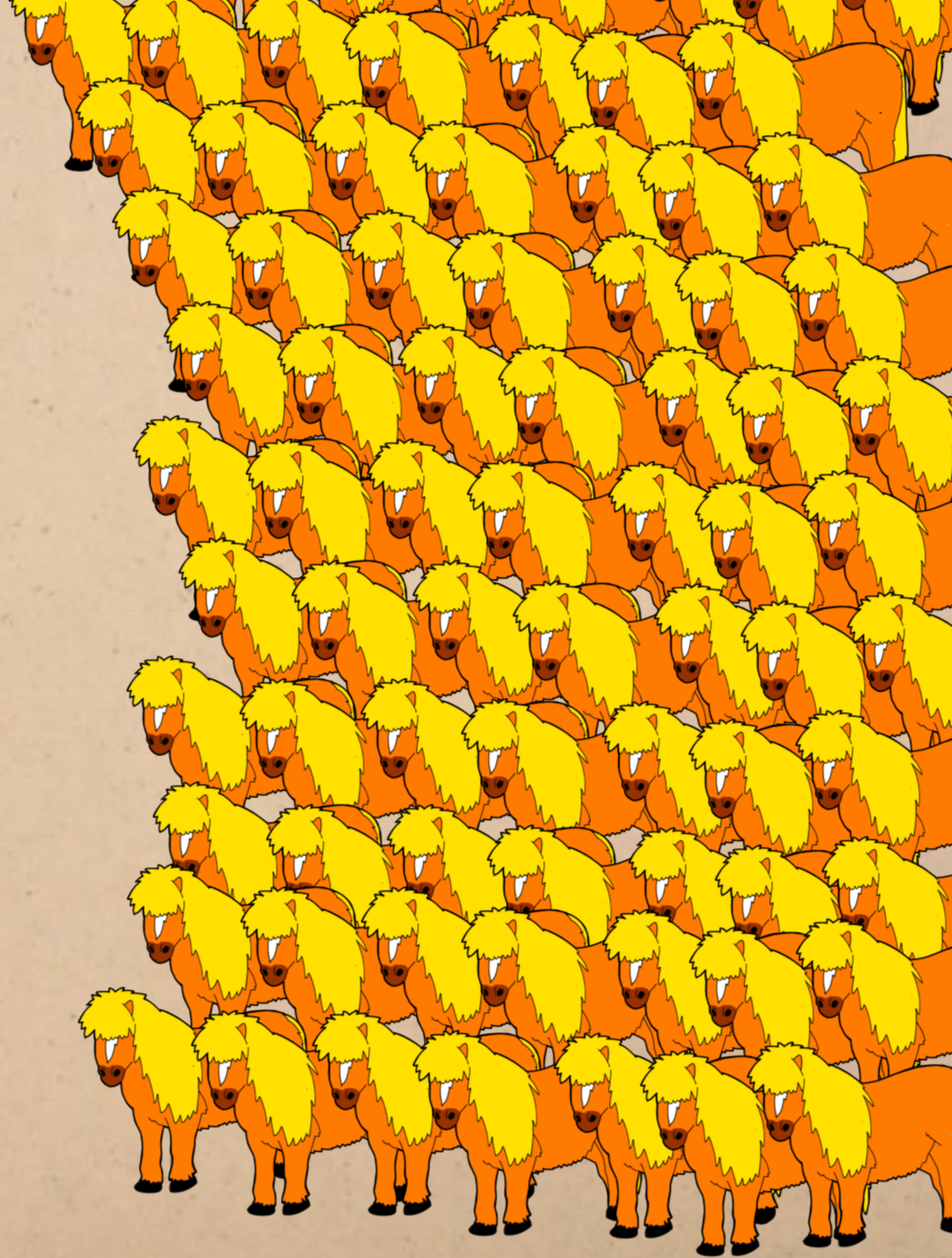


I don't enjoy the idea of subjecting 100 cute Li'l Sebastians to anything more violent than a slightly over-enthusiastic hug, but if it was a matter of survival, I'd rather face them than a normal horse-sized anything...



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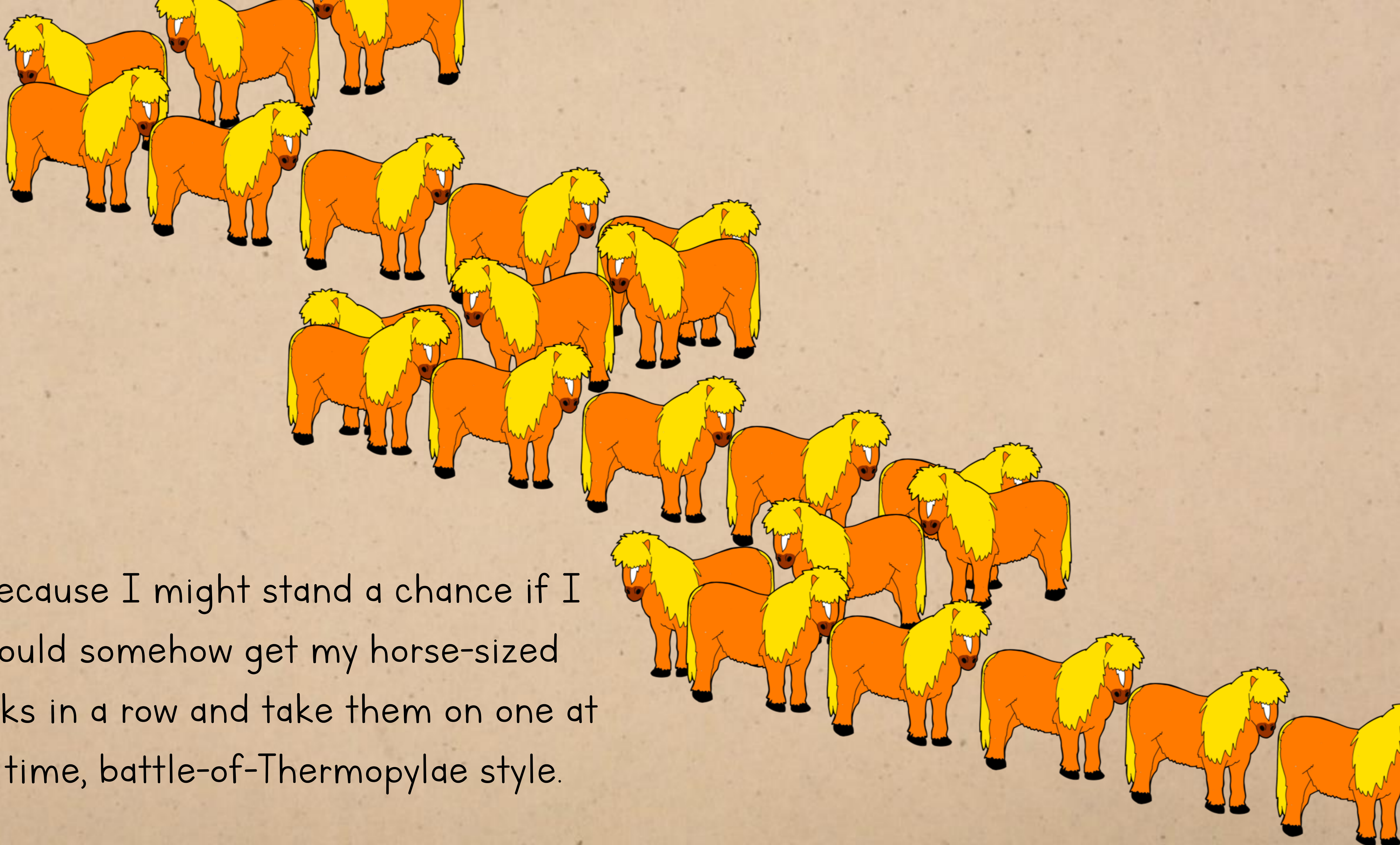
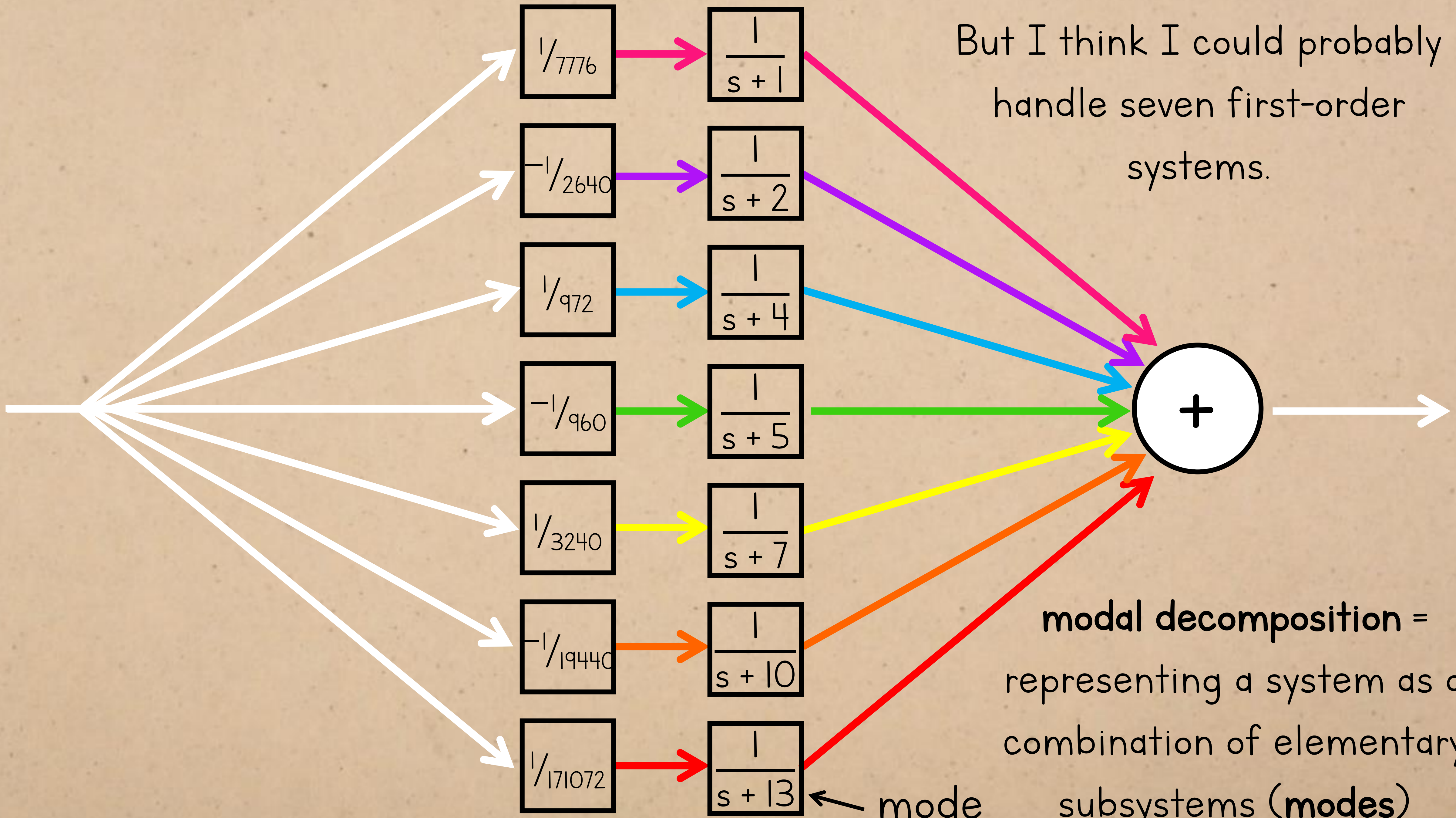


...because I might stand a chance if I could somehow get my horse-sized ducks in a row and take them on one at a time, battle-of-Thermopylae style.

And likewise, I wouldn't want to deal with a seventh-order system...

$$\boxed{\frac{1}{s^7 - 42s^6 + 700s^5 - 5950s^4 + 27559s^3 - 68488s^2 + 82620s - 36400}}$$



But I think I could probably handle seven first-order systems.

modal decomposition = representing a system as a combination of elementary subsystems (**modes**)

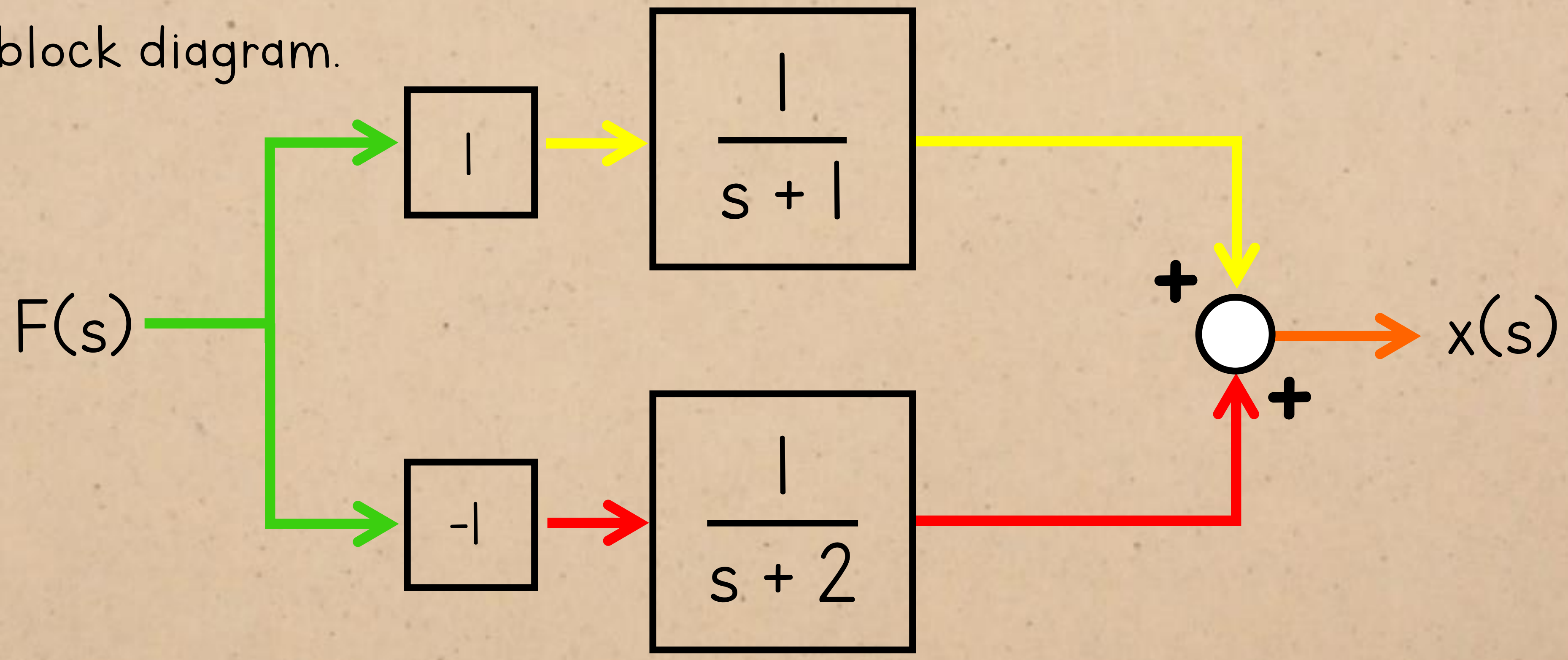
DYNAMIC RESPONSE

Having decomposed the response into these modes...

$$x(s) = \frac{1}{s+1}F(s) - \frac{1}{s+2}F(s)$$

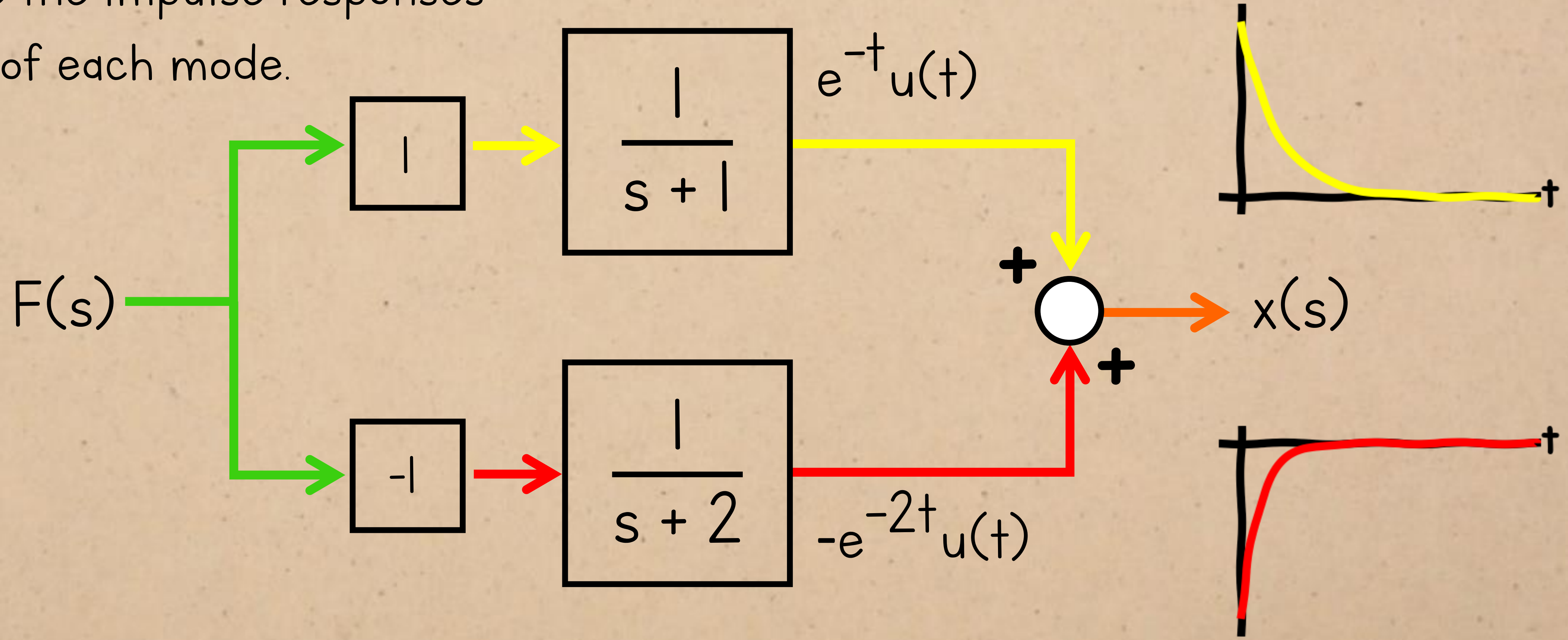
DYNAMIC RESPONSE

...we can visualize it using this block diagram.

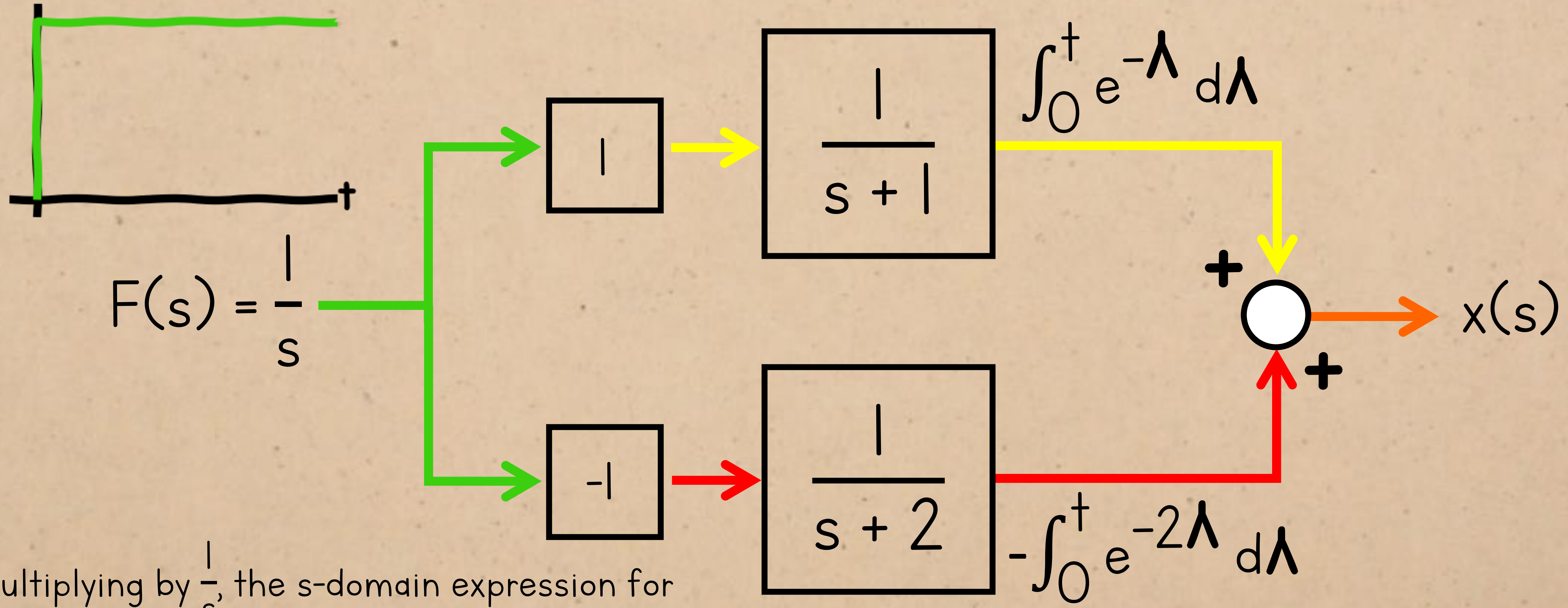


DYNAMIC RESPONSE

These are the impulse responses of each mode.

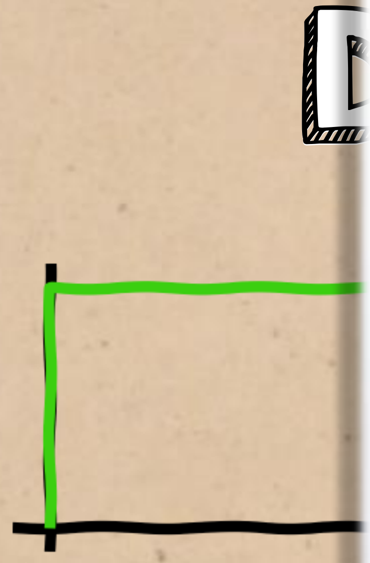


DYNAMIC RESPONSE



Multiplying by $\frac{1}{s}$, the s-domain expression for the unit step, is the same as integrating in the time-domain.

So the step response is the integral of the impulse response. (We also learned this when we covered the integration property of convolution last year.)



F(s) =

Linearity	$ax(t) + bv(t) \Longleftrightarrow aX(s) + bV(s)$
Time shift	$x(t-a)u(t-a) \Longleftrightarrow e^{-as}X(s) \quad a \geq 0$
Time scaling	$x(at) \Longleftrightarrow \frac{1}{a}X(\frac{s}{a}) \quad a > 0$
Frequency differentiation	$t^n x(t) \Longleftrightarrow (-1)^n X^{(n)}(s)$
Frequency shift	$e^{at}x(t) \Longleftrightarrow X(s-a)$
Differentiation	$x'(t) \Longleftrightarrow sX(s) - x(0^-)$
	$x''(t) \Longleftrightarrow s^2X(s) - sx(0^-) - x'(0^-)$
	$x^{(n)}(t) \Longleftrightarrow s^nX(s) - s^{n-1}x(0^-) - \cdots - x^{(n-1)}(0^-)$
Integration	$\int_{0^-}^{t} x(\lambda)d\lambda \Longleftrightarrow \frac{1}{s}X(s)$
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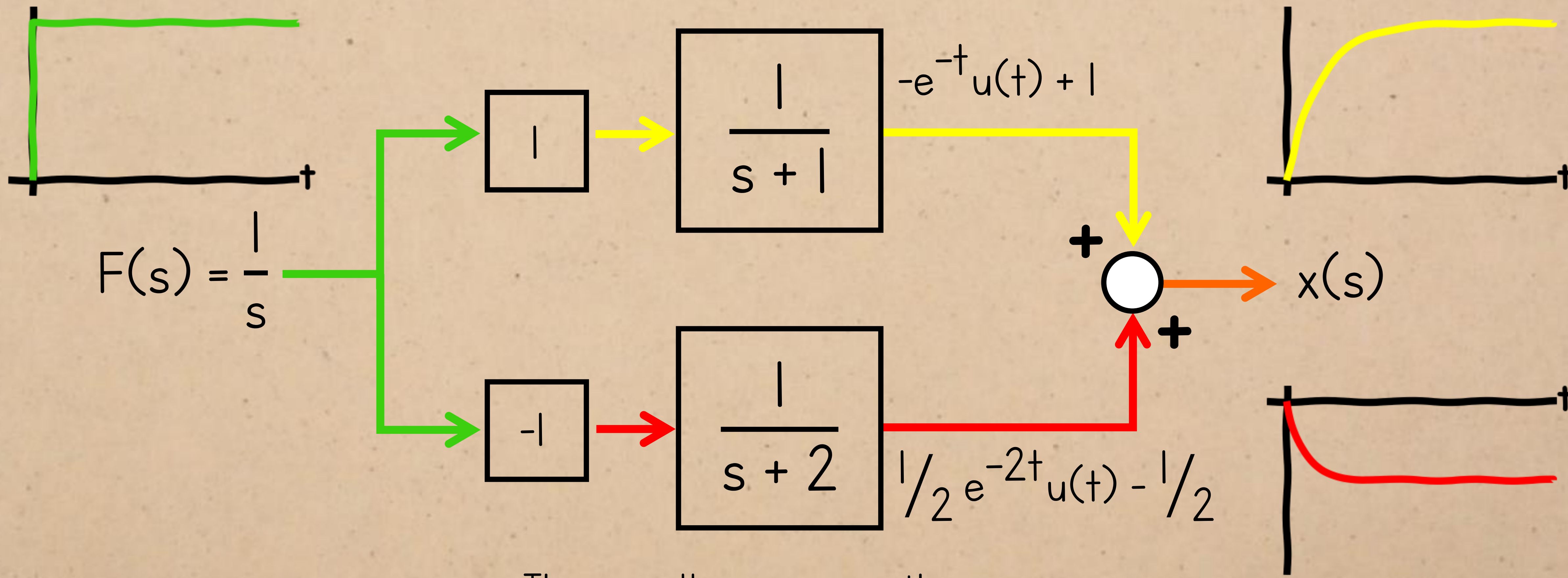
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
$tu(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$e^{\lambda t}u(t)$	$\frac{1}{s-\lambda}$
$te^{\lambda t}u(t)$	$\frac{1}{(s-\lambda)^2}$
$t^n e^{\lambda t}u(t)$	$\frac{n!}{(s-\lambda)^{n+1}}$
$\cos(bt)u(t)$	$\frac{s}{s^2+b^2}$

DYNAMIC RESPONSE



The overall response is the sum of the responses to the modes.

DYNAMIC RESPONSE

$$x(t) = e^{-t}u(t) - \frac{1}{2}e^{-2t}u(t) + \frac{1}{2}$$

DYNAMIC RESPONSE

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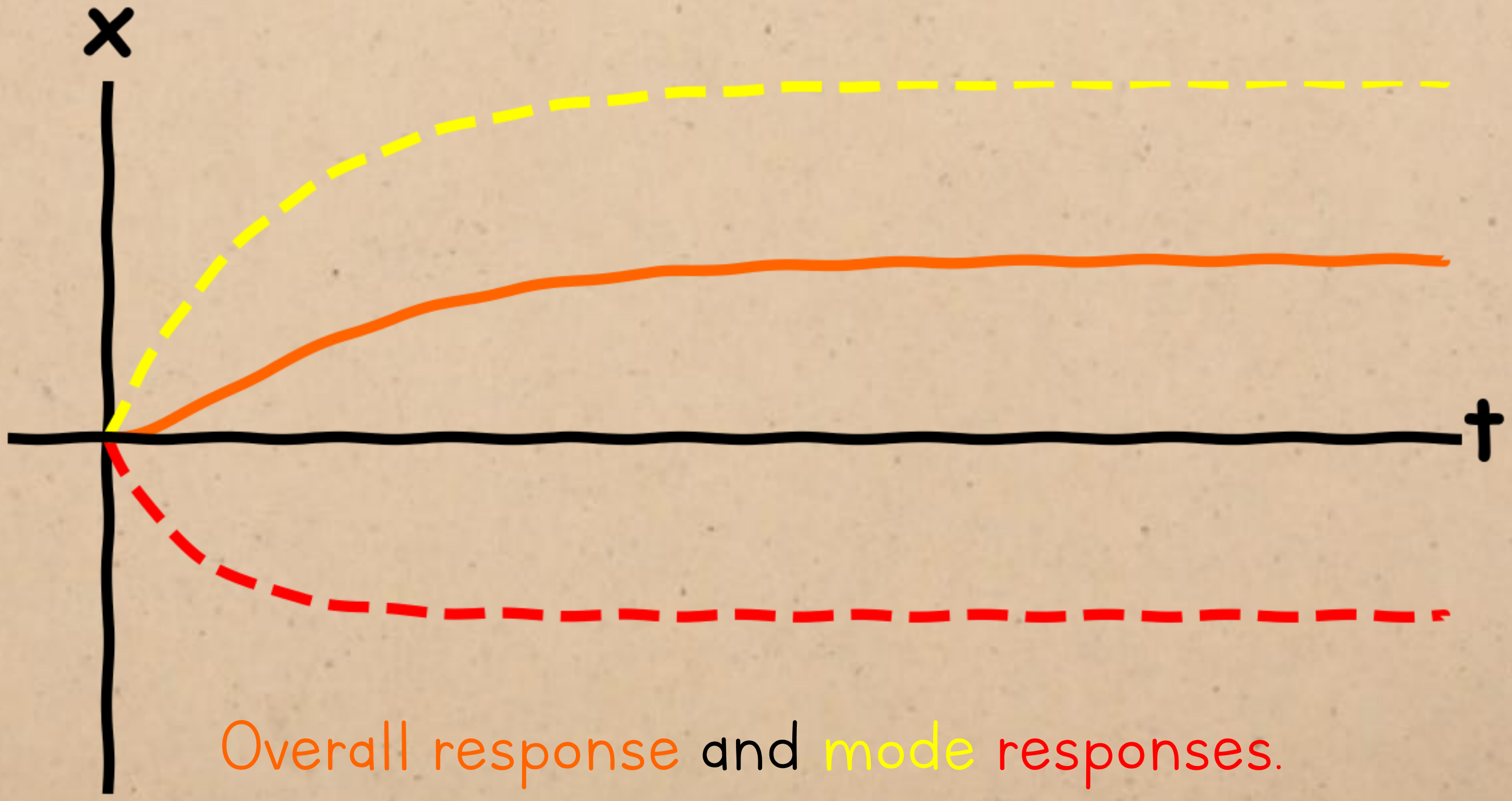
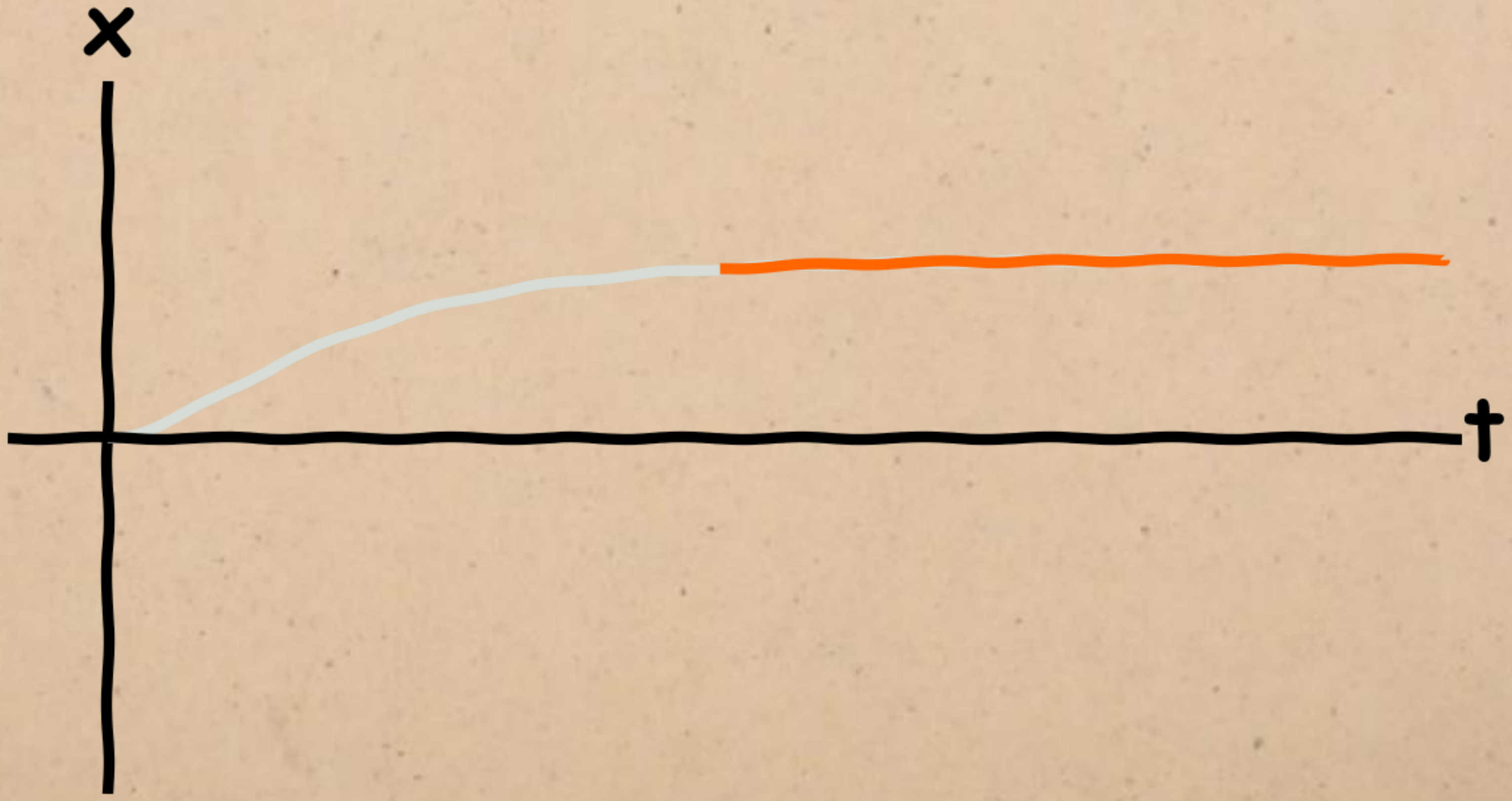


Overall response and mode responses.

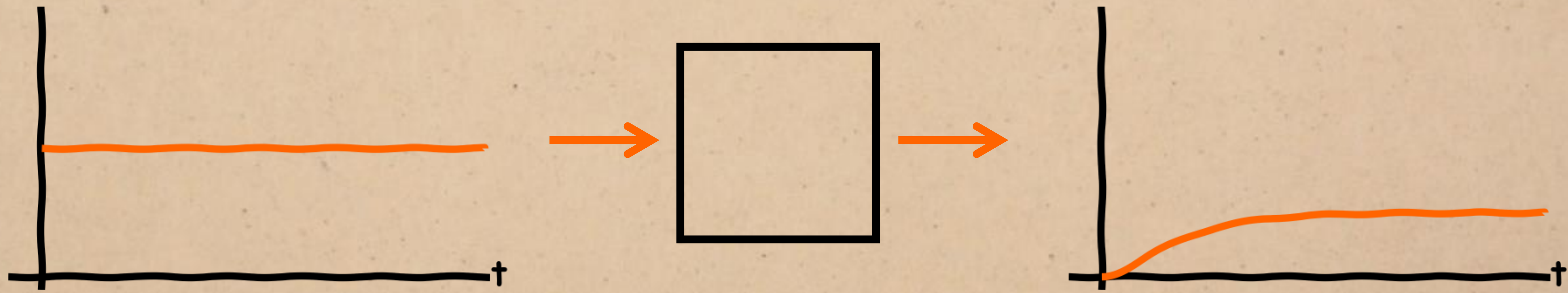
STEADY-STATE RESPONSE

steady state = part of the response that remains as time tends to infinity.



STEADY-STATE RESPONSE

The form of the steady state depends on the input:



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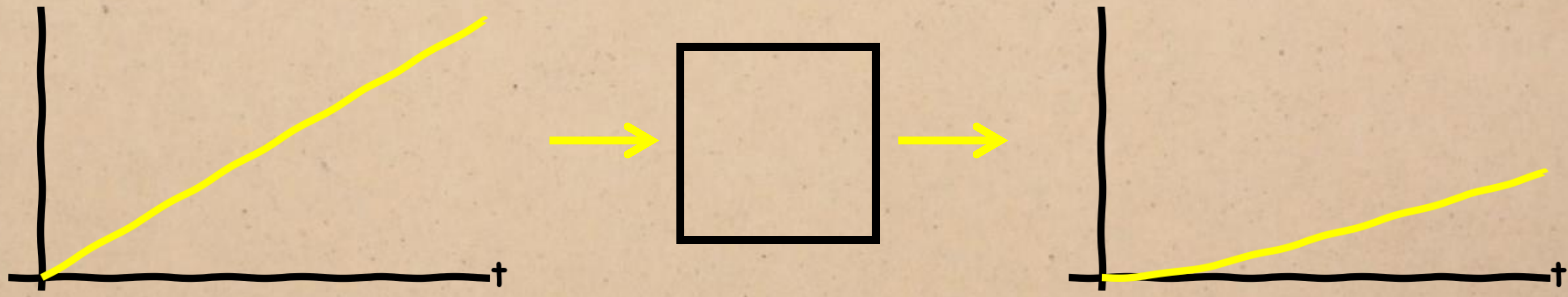

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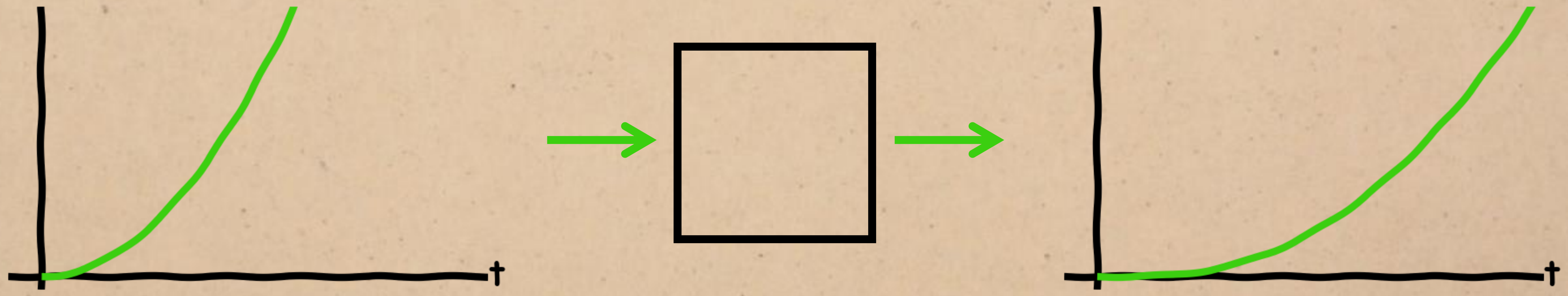

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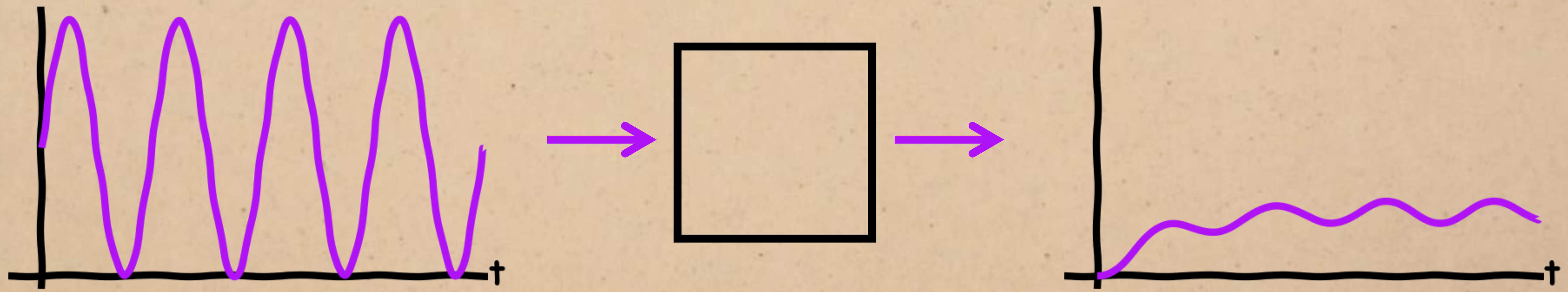
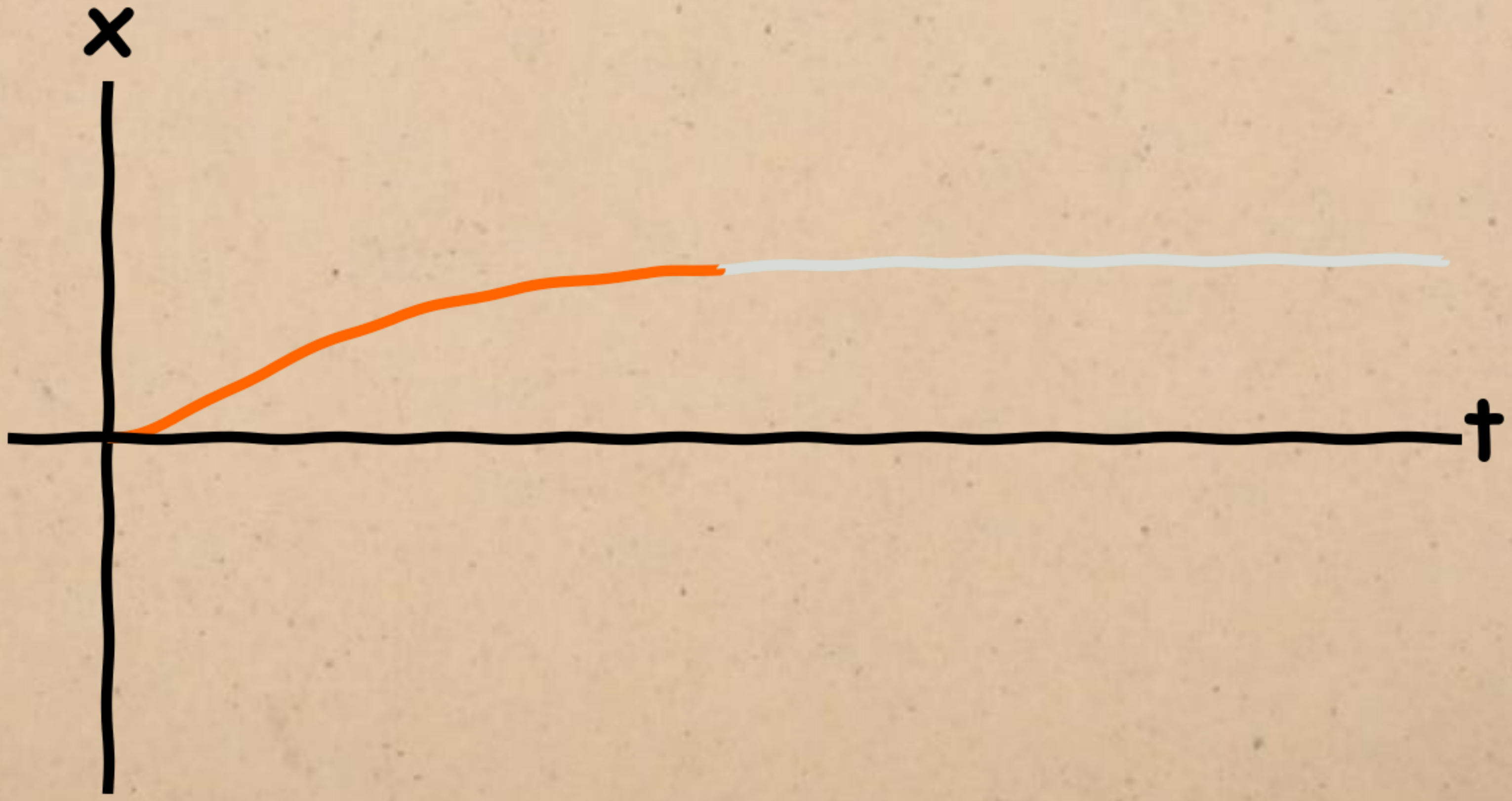


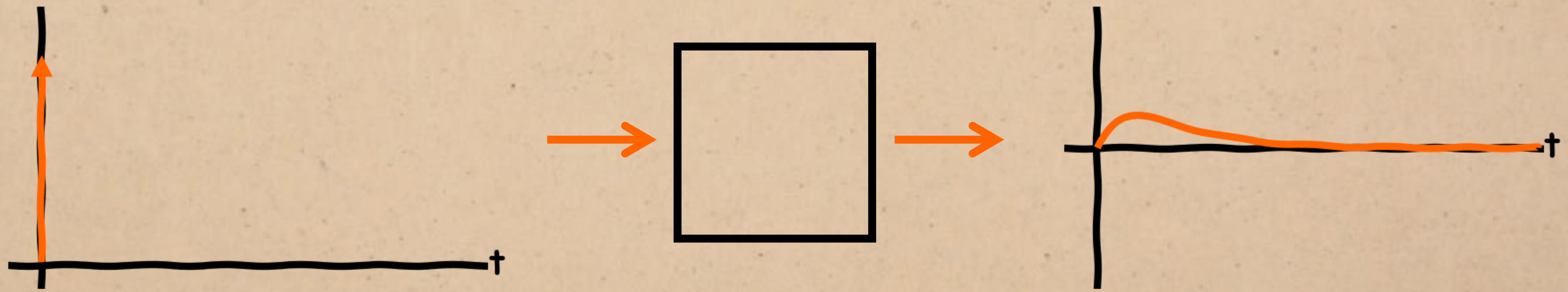
TRANSIENT RESPONSE

transient response = part of the response that decays away over time



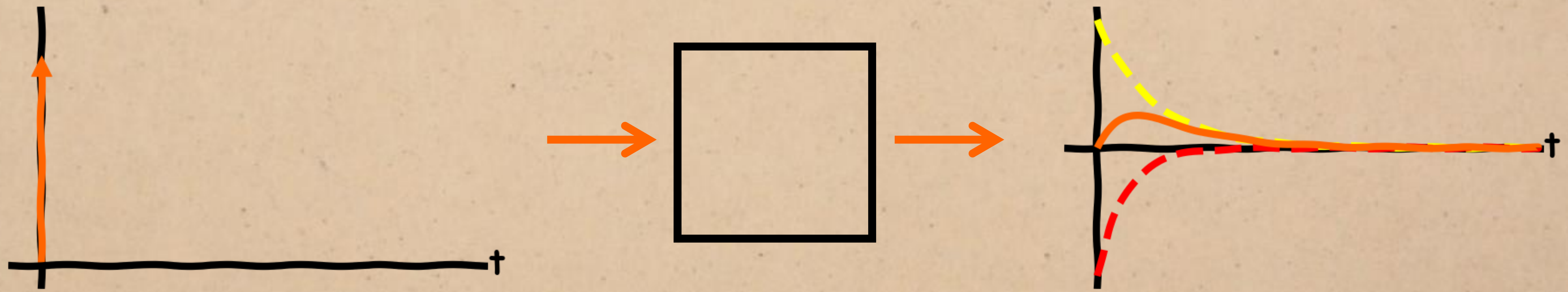
TRANSIENT RESPONSE

transient response = impulse response



TRANSIENT RESPONSE

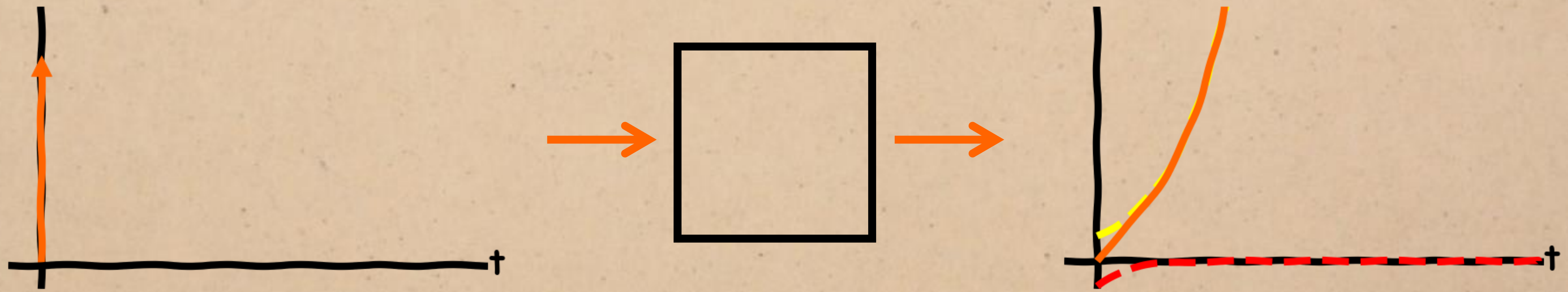
The transient response dies away because all the exponential modes are decaying.



TRANSIENT RESPONSE

If any of the modes are growing, the transient response will also grow to infinity.

= unstable system



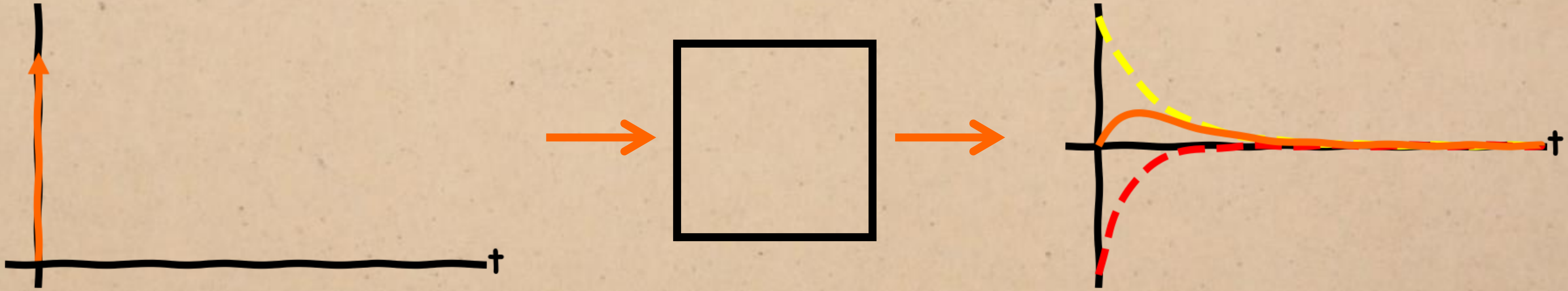
BIBO = Bounded input, Bounded output

This way of defining stability states that the system is stable if a bounded (i.e. non-infinite) response is guaranteed to produce a bounded output. Here, the system is unstable because the bounded impulse lead to a response that grows infinitely.

TRANSIENT RESPONSE

The sign of number in the time-domain response determines whether the system is stable or unstable.

$$x(t) = e^{-1t}u(t) - e^{-2t}u(t)$$



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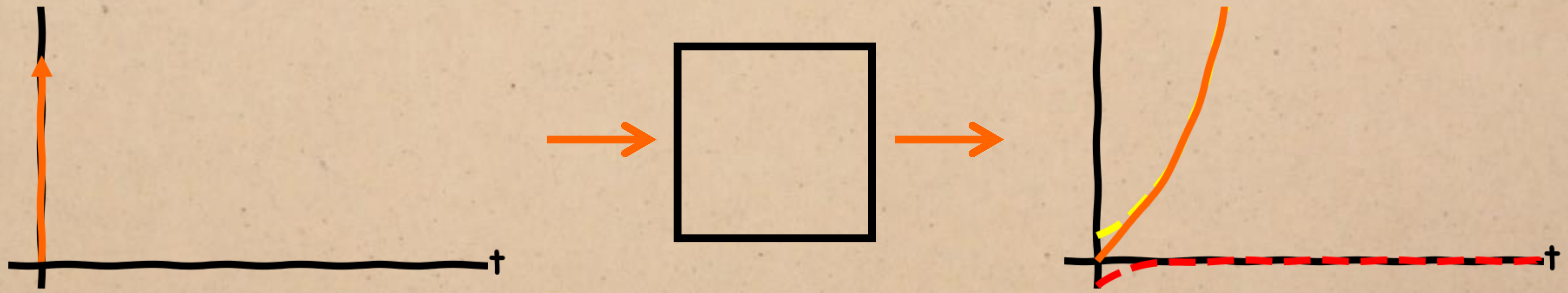
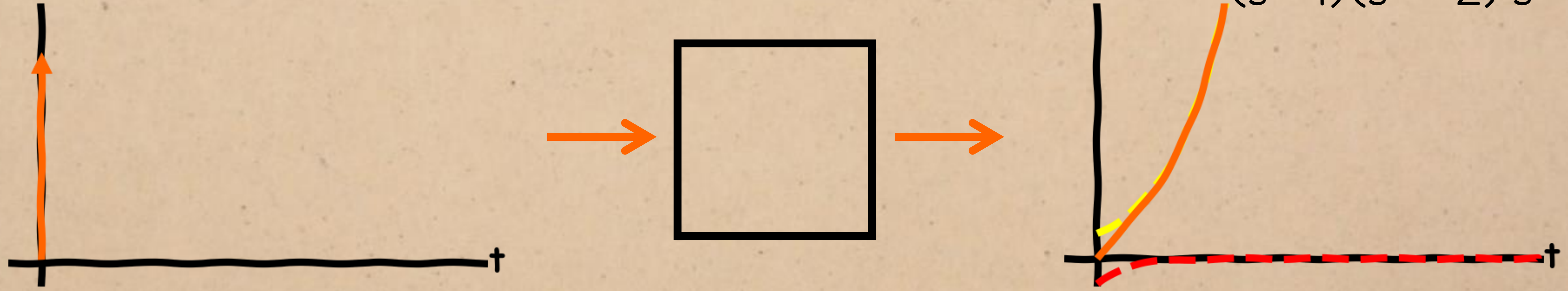


TRANSIENT RESPONSE

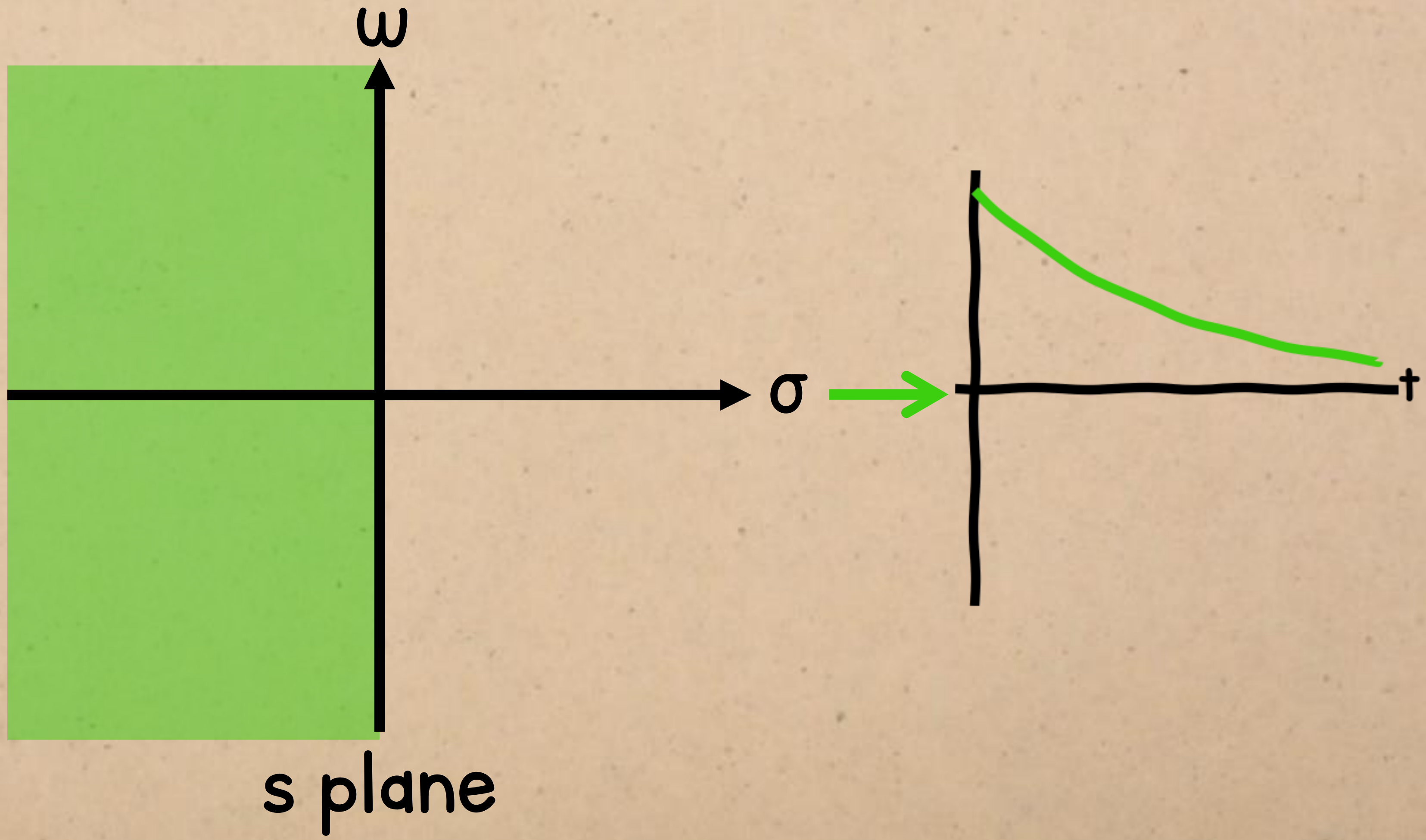
This means that the locations of the poles determine the stability of the system.

$$x(s) = \frac{1}{(s-1)(s--2)}\,\frac{1}{s}$$



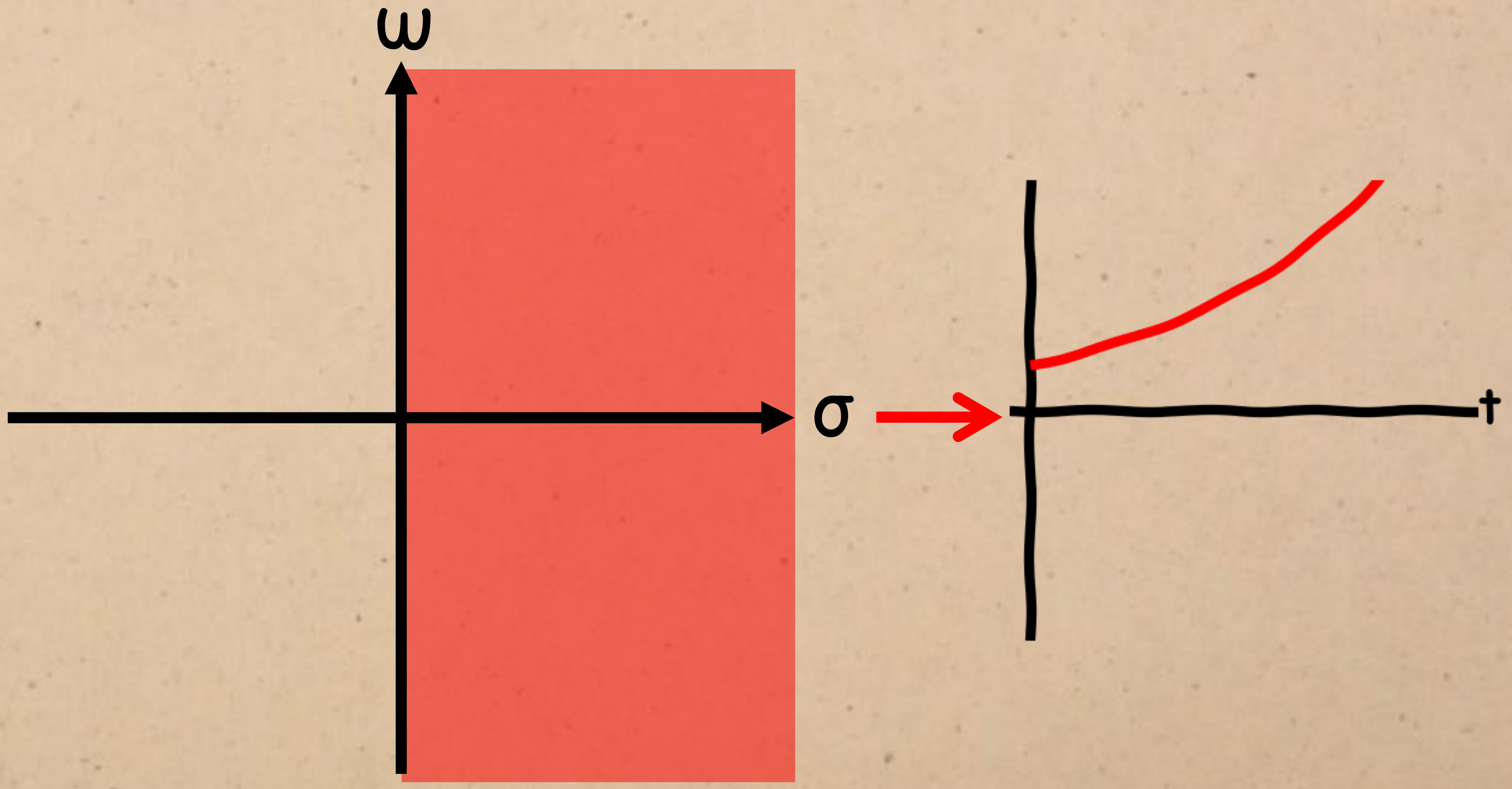
TRANSIENT RESPONSE

Poles in the **left-half plane** (negative poles) are **stable**.



TRANSIENT RESPONSE

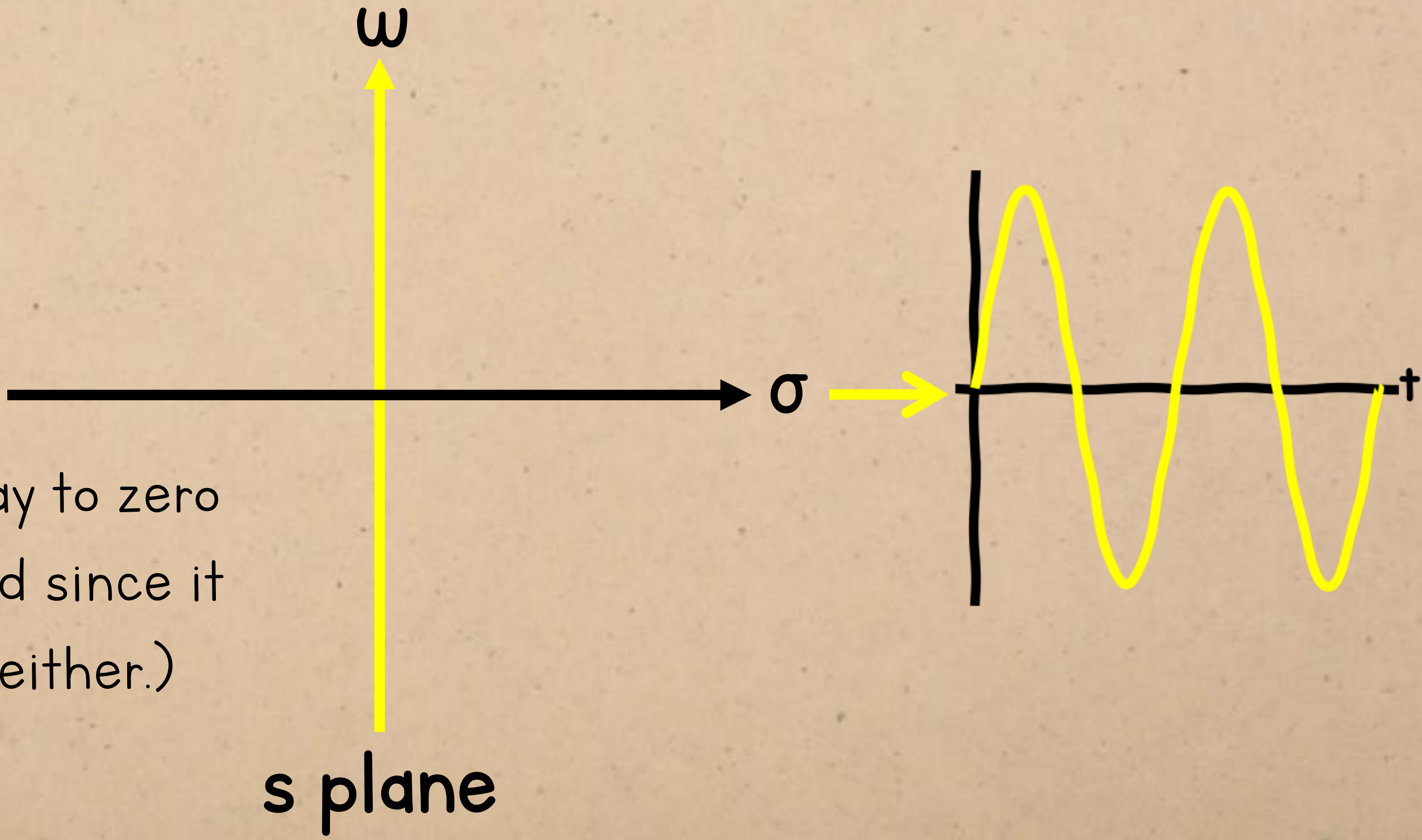
Poles in the **right-half plane** (positive poles) are **unstable.**



s plane

TRANSIENT RESPONSE

Poles on the **axis** (real part = 0) are **marginally stable**.

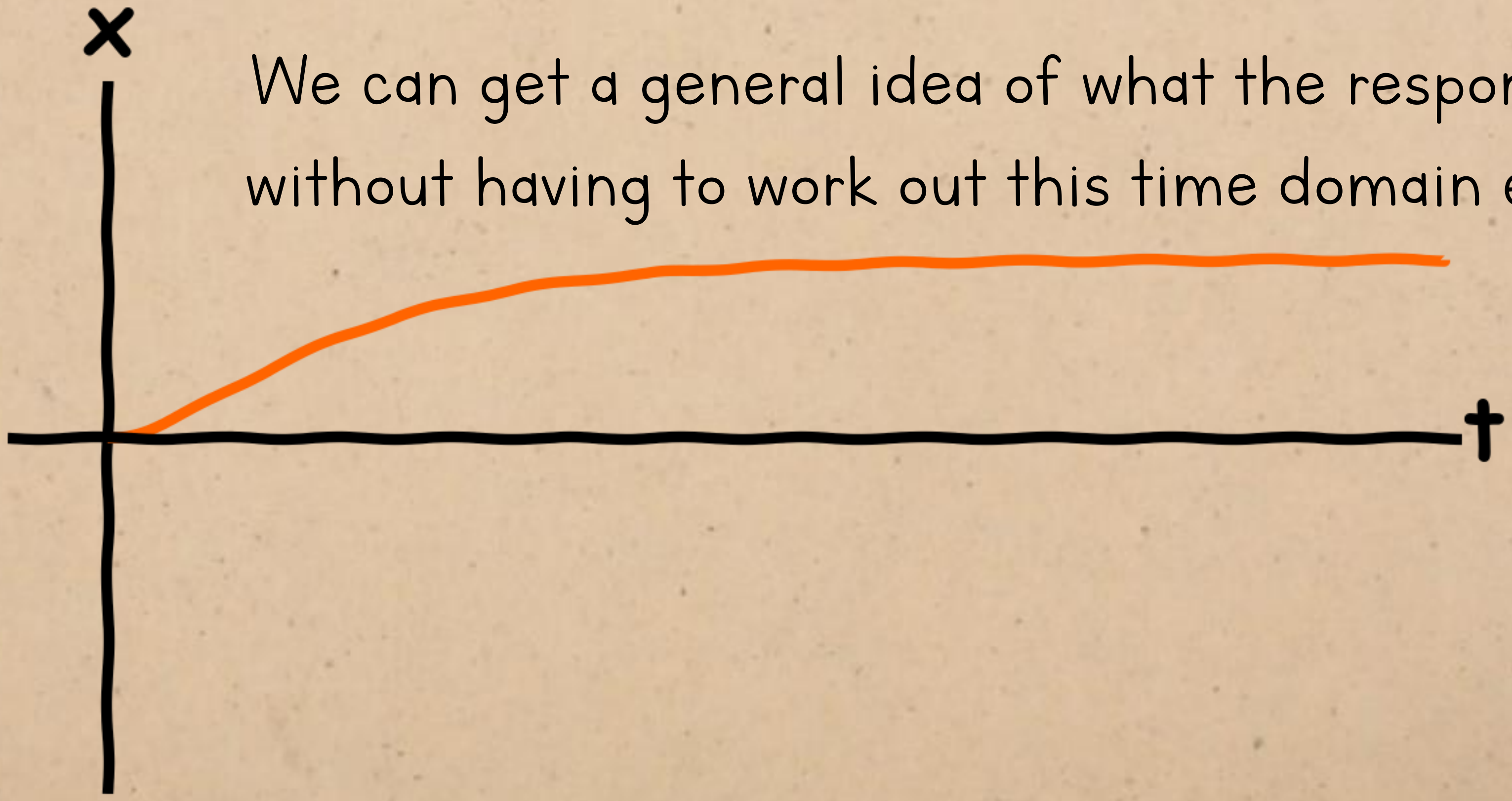


(The sinusoid doesn't decay to zero over time, but it is bounded since it isn't growing to infinity, either.)

TRANSIENT RESPONSE

$$x(t) = e^{-t}u(t) - \frac{1}{2}e^{-2t}u(t) + \frac{1}{2}$$

We can get a general idea of what the response is doing without having to work out this time domain expression....



TRANSIENT RESPONSE

$$x(s) = \frac{1}{(s+1)(s+2)}\,\frac{1}{s}$$

initial value theorem (IVT):

$$x(0) = \lim_{s \to \infty} s x(s)$$

final value theorem (FVT):

$$x(\infty) = \lim_{s \to 0} s x(s)$$

The initial and final value theorems can tell you the value of the time-domain response at $t = 0$, and as t tends to infinity, using only the transfer function

TRANSIENT RESPONSE

$$x(s) = \frac{1}{(s+1)(s+2)}\,\frac{1}{s}$$

initial value theorem (IVT):

$$x(0) = \lim_{s \to \infty} sx(s)$$

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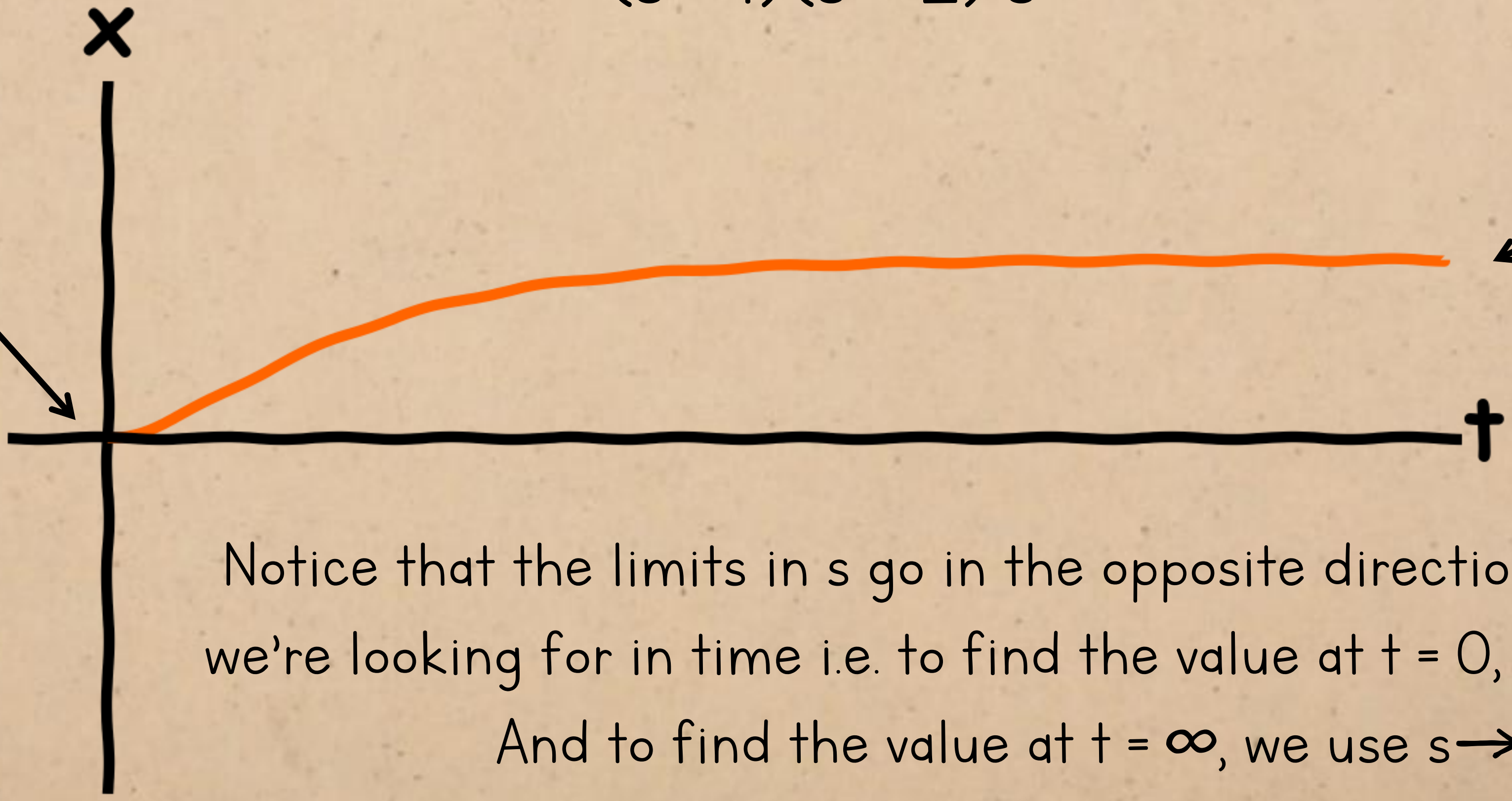


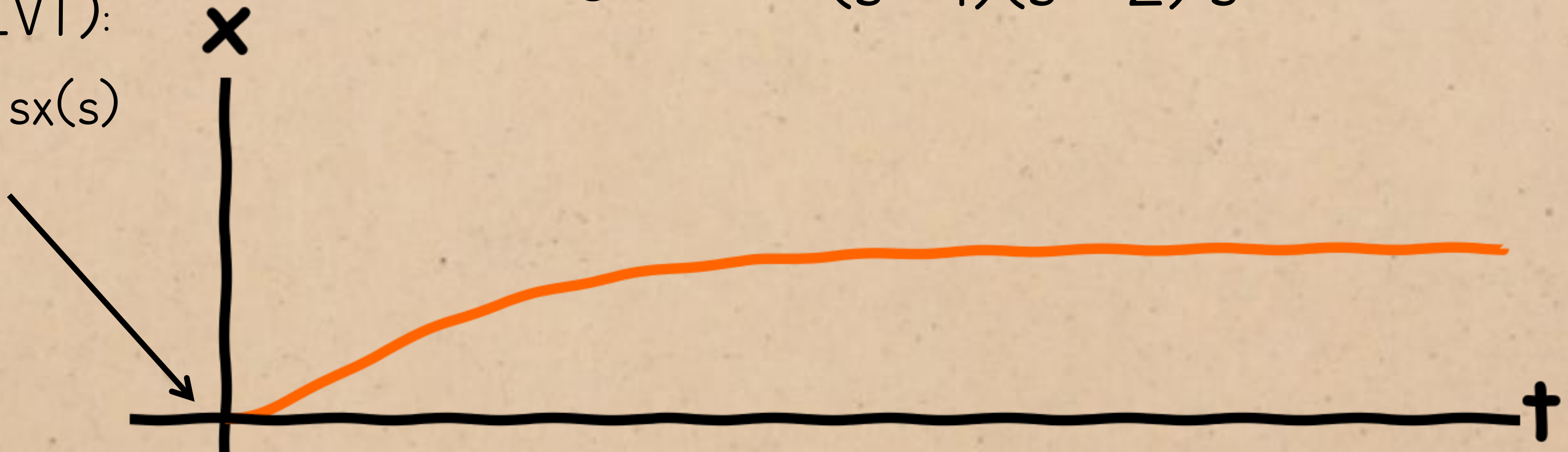
Notice that the limits in s go in the opposite direction to the value we're looking for in time i.e. to find the value at t = 0, we use s $\rightarrow$0, And to find the value at t = ∞, we use s$\rightarrow$0.

TRANSIENT RESPONSE

initial value theorem (IVT):

$$x(0) = \lim_{s \to \infty} sx(s)$$

$$x(0) = \lim_{s \to \infty} s \frac{1}{(s+1)(s+2)} \frac{1}{s} = 0$$



If we apply the initial value theorem, we get a value of 0, which agrees with the plot.

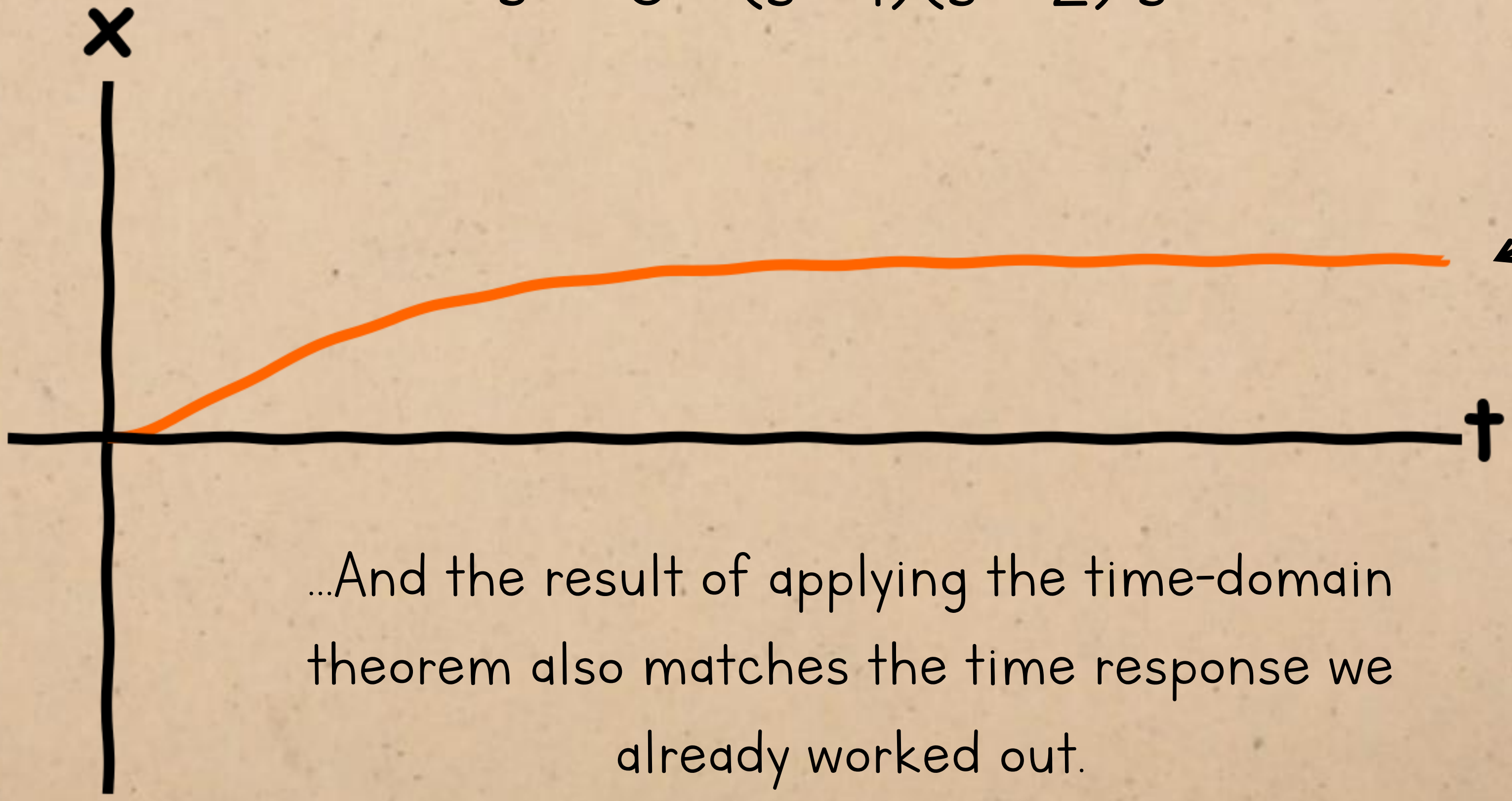
TRANSIENT RESPONSE

$$x(\infty) = \lim_{s \to 0} s \frac{1}{(s+1)(s+2)} \frac{1}{s} = \frac{1}{2}$$

final value theorem (FVT):

$$x(\infty) = \lim_{s \to 0} sx(s)$$

$$= 0.5$$



...And the result of applying the time-domain theorem also matches the time response we already worked out.

FREE RESPONSE

We probably won't consider the free response all that often in the course, because we'll usually assume initial rest, but it's still worth knowing about.

$$x(s) = \frac{1}{(s+1)(s+2)} F(s) + \frac{(3+s)x(0) + \dot{x}(0)}{(s+1)(s+2)}$$

FREE RESPONSE

$$x(s) = \frac{(3 + s)x(0) + \dot{x}(0)}{(s + 1)(s + 2)}$$

FREE RESPONSE

Even though there's an 's' in the numerator...

$$x(s) = \frac{(3 + s)x(0) + \dot{x}(0)}{(s + 1)(s + 2)}$$

FREE RESPONSE

We still end up with two terms with constant denominators (determined by the initial conditions) when we do the partial fraction expansion.

$$x(s) = \frac{(3 + s)x(0) + \dot{x}(0)}{(s + 1)(s + 2)}$$

$$\frac{A}{s+1} + \frac{B}{s+2} = \frac{(3 + s)x(0) + \dot{x}(0)}{(s + 1)(s + 2)} \therefore A(s + 2) + B(s + 1) = (3 + s)x(0) + \dot{x}(0)$$

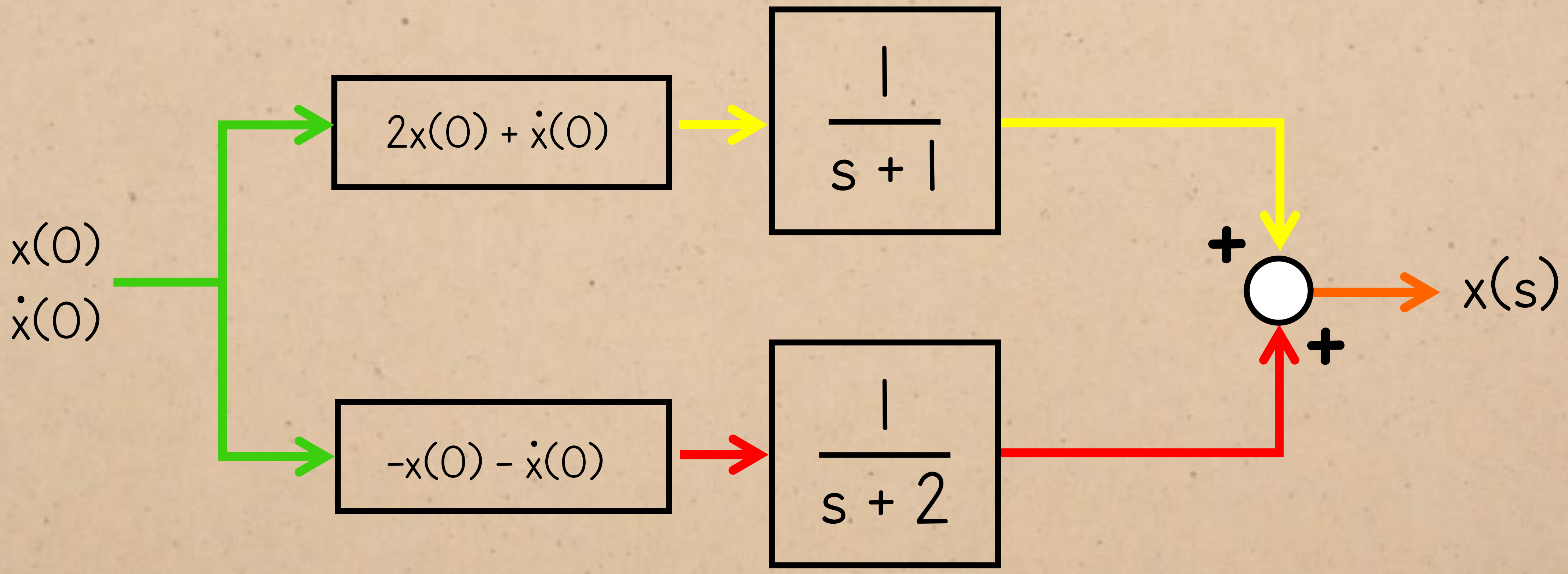
let $s = -1$: $A(-1 + 2) + B(-1 + 1) = (3 + -1)x(0) + \dot{x}(0) \therefore A = 2x(0) + \dot{x}(0)$

let $s = -2$: $A(-2 + 2) + B(-2 + 1) = (3 + -2)x(0) + \dot{x}(0) \therefore -B = x(0) + \dot{x}(0) \therefore B = -x(0) - \dot{x}(0)$

$$\therefore \frac{2x(0) + \dot{x}(0)}{s + 1} - \frac{x(0) + \dot{x}(0)}{s + 2}$$

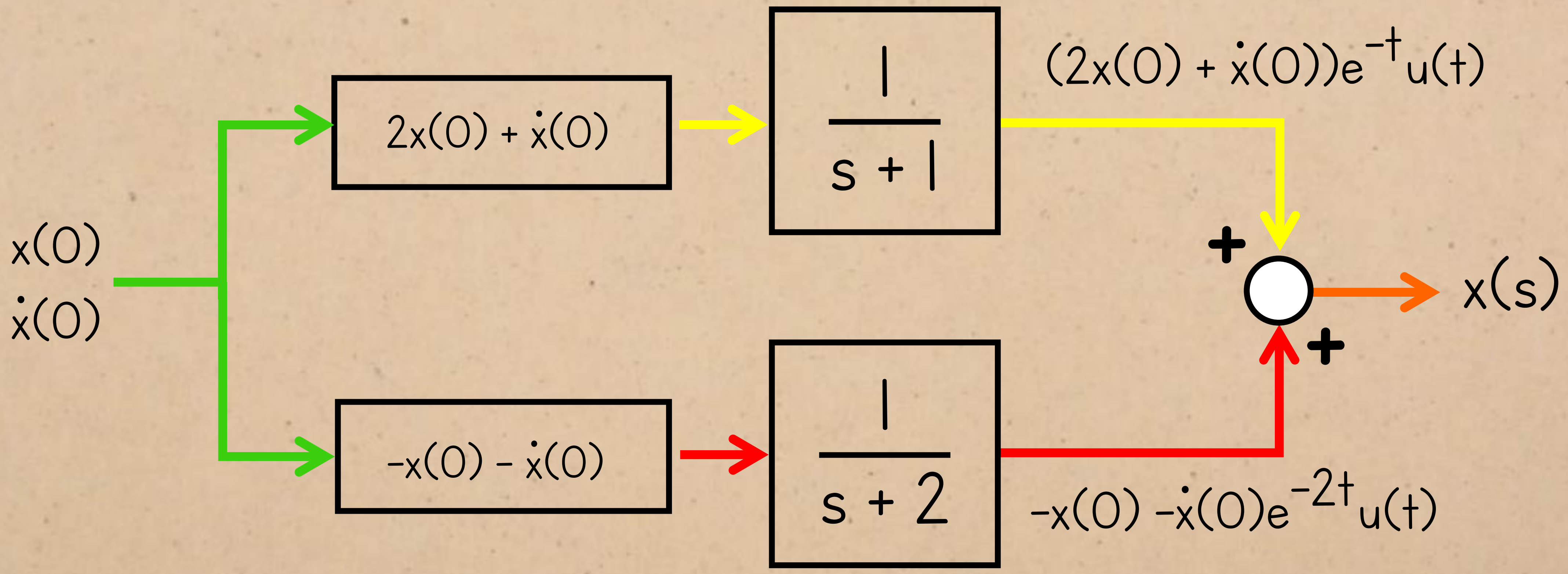
DYNAMIC RESPONSE

So this is what the modal decomposition looks like as a block diagram.



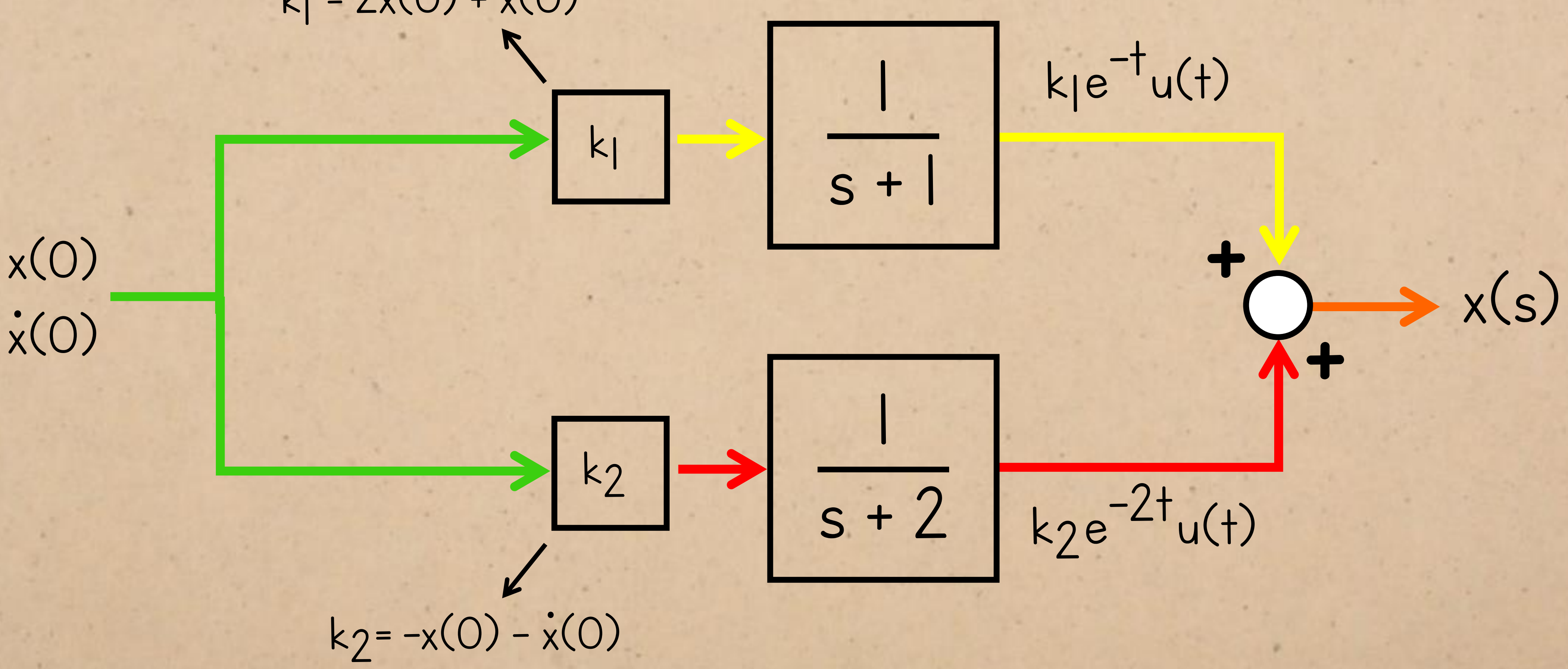
DYNAMIC RESPONSE

The impulse responses:



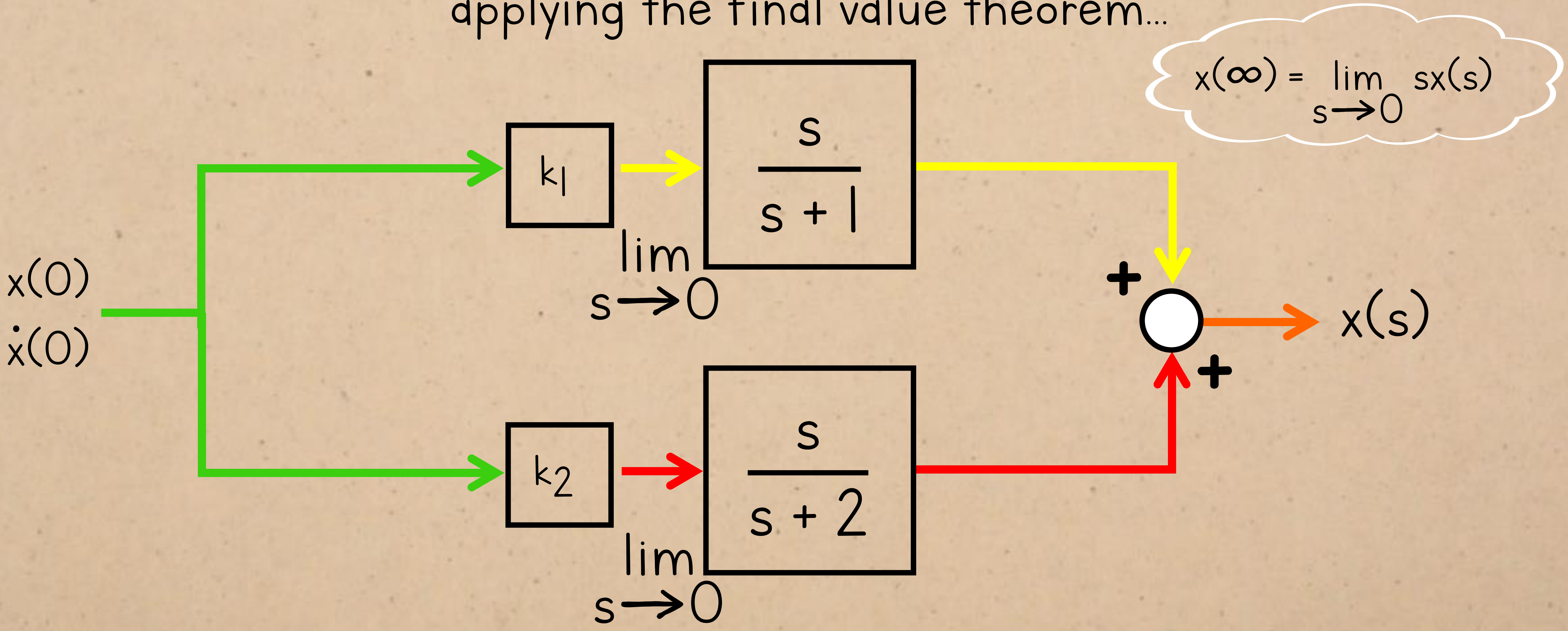
DYNAMIC RESPONSE

Let's call these values k_1 and k_2 for neatness:



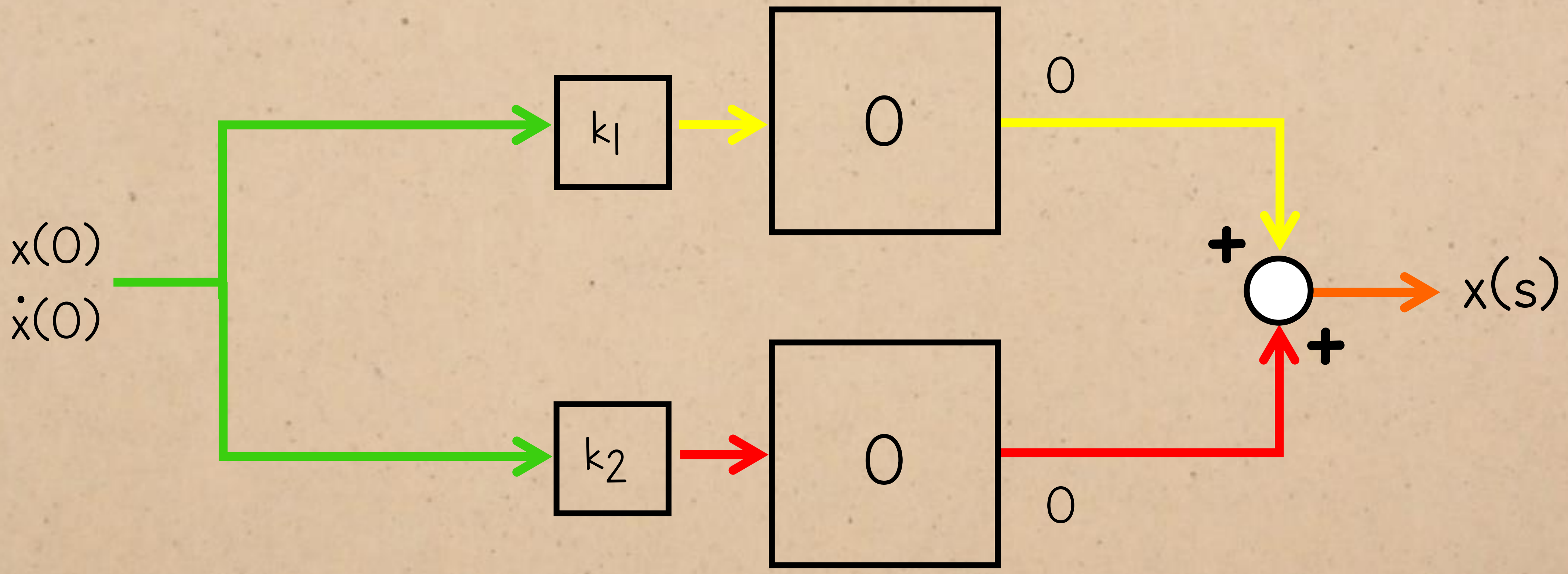
DYNAMIC RESPONSE

applying the final value theorem...



DYNAMIC RESPONSE

applying the final value theorem...

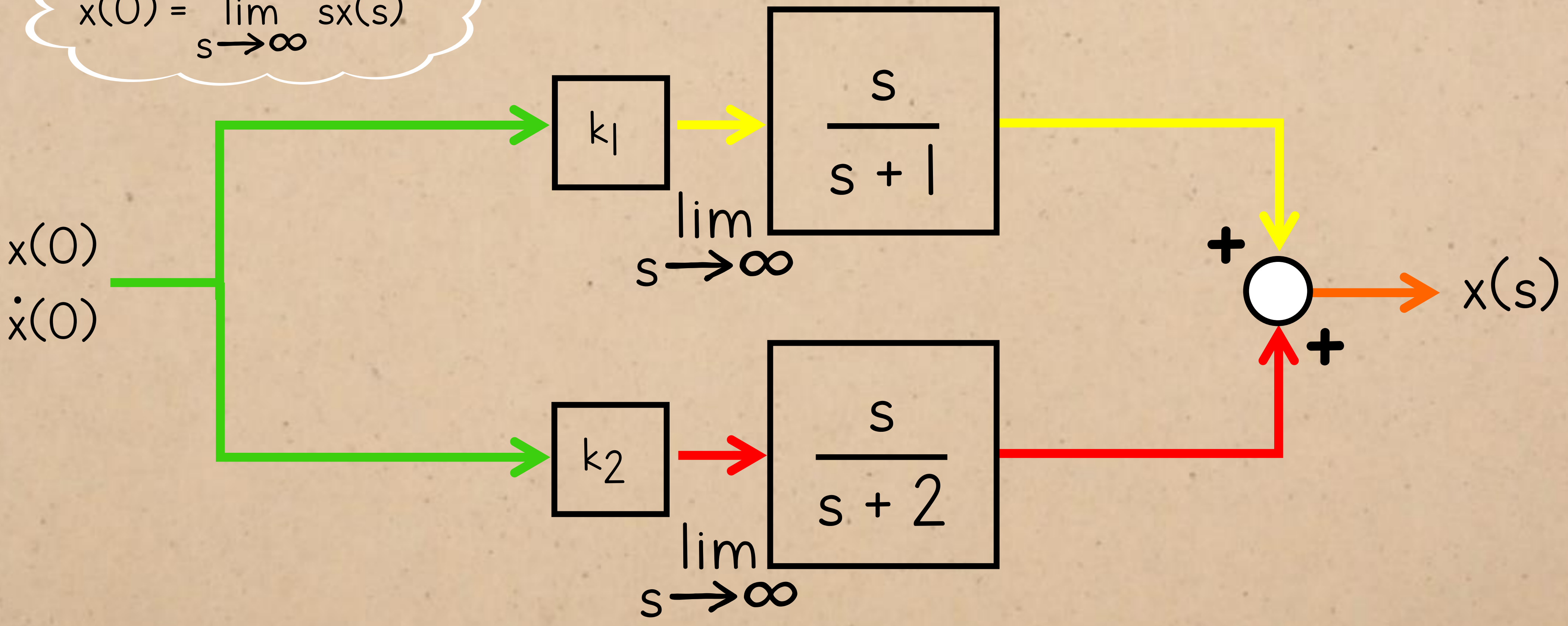


...so, the free response decays to zero regardless of the initial conditions.

DYNAMIC RESPONSE

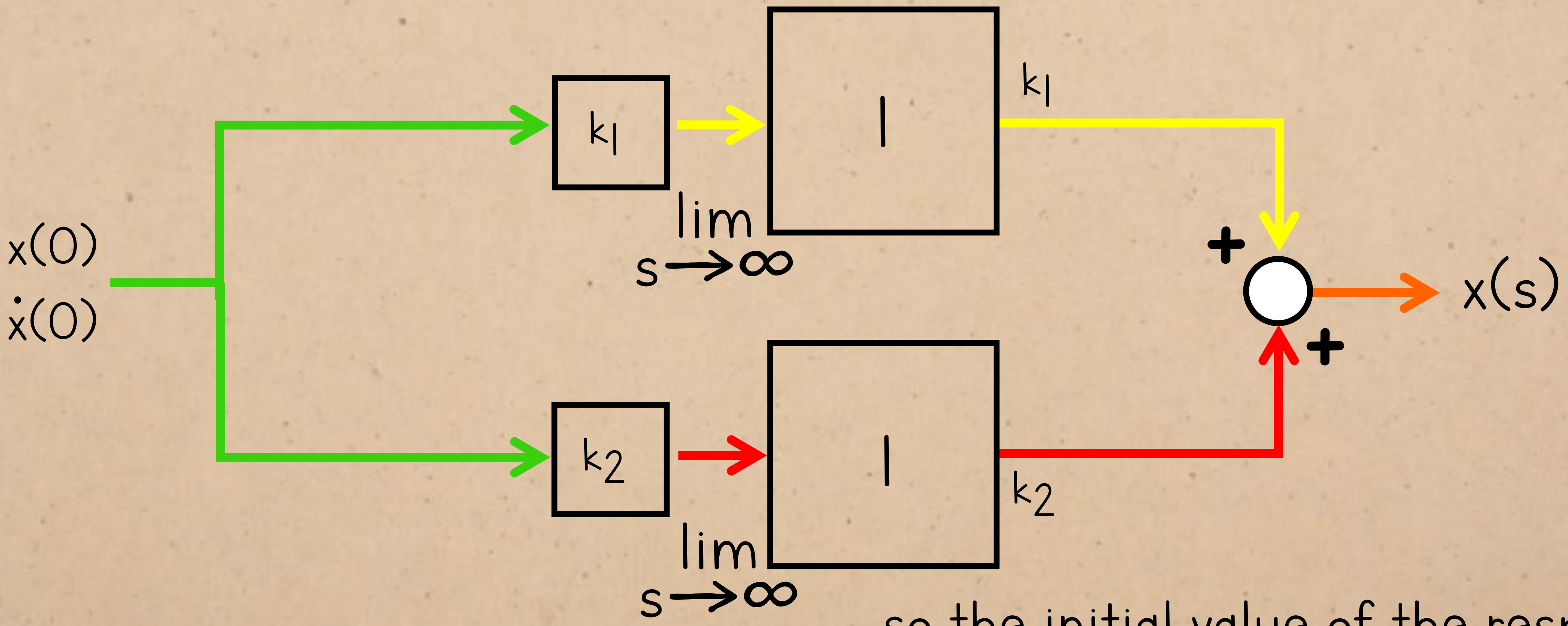
applying the initial value theorem...

$$x(0) = \lim_{s \to \infty} sx(s)$$



DYNAMIC RESPONSE

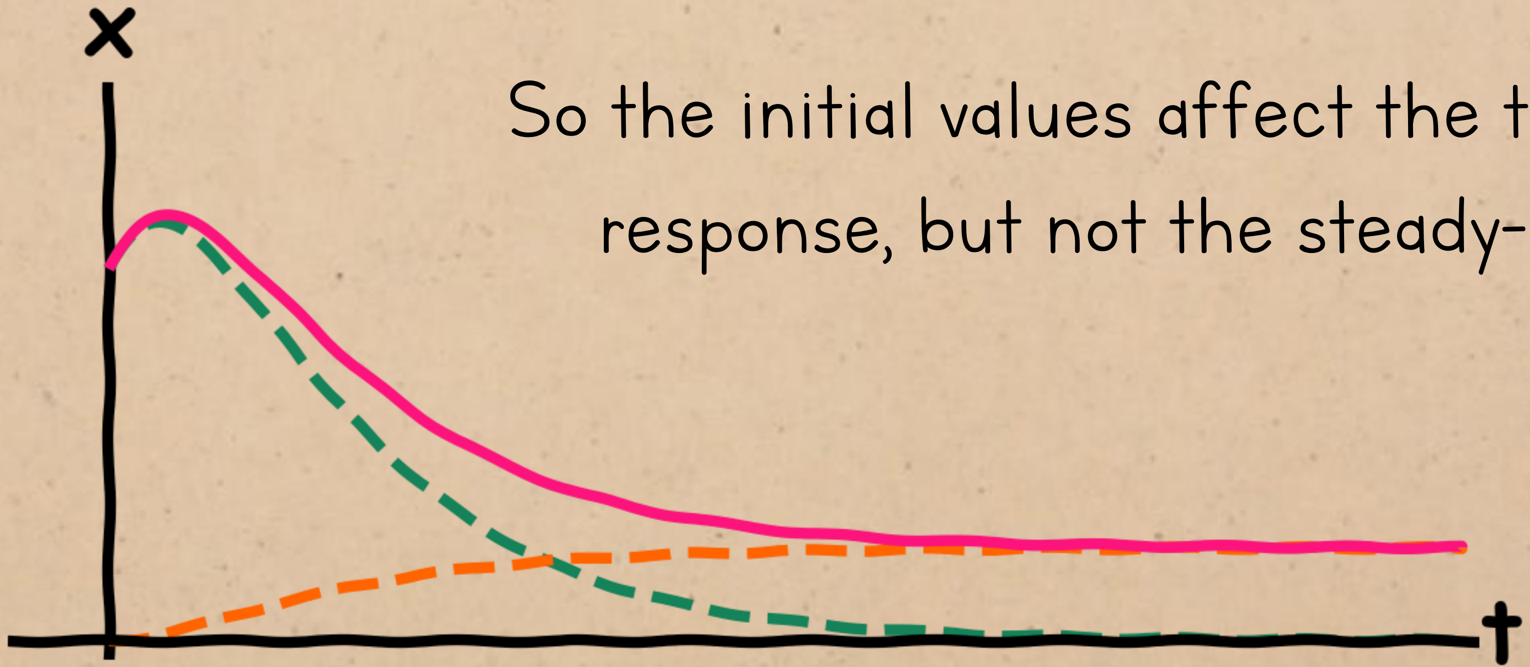
applying the initial value theorem...



...so the initial value of the response is affected by the initial conditions. (duh)

FREE RESPONSE

complete response = forced response + free response

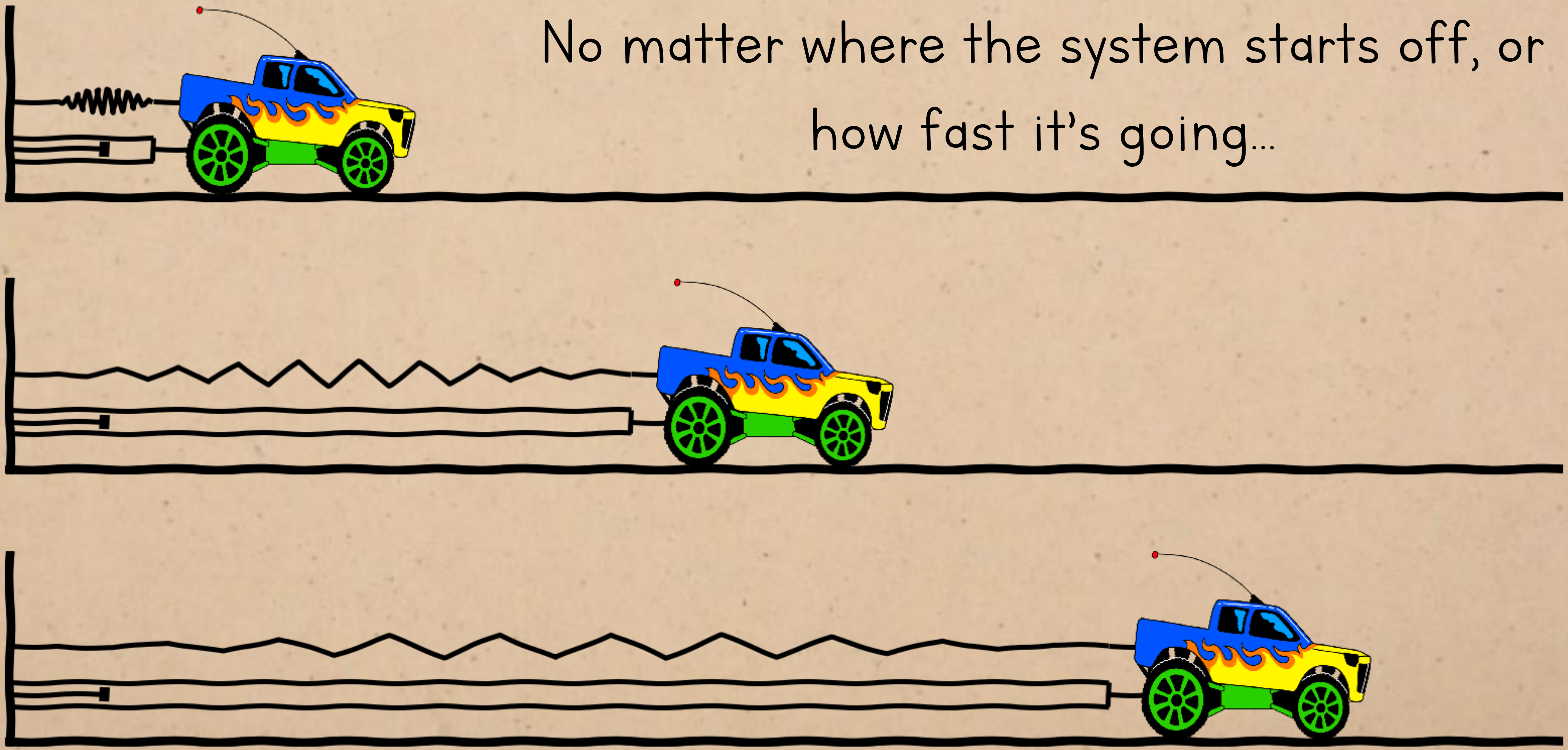


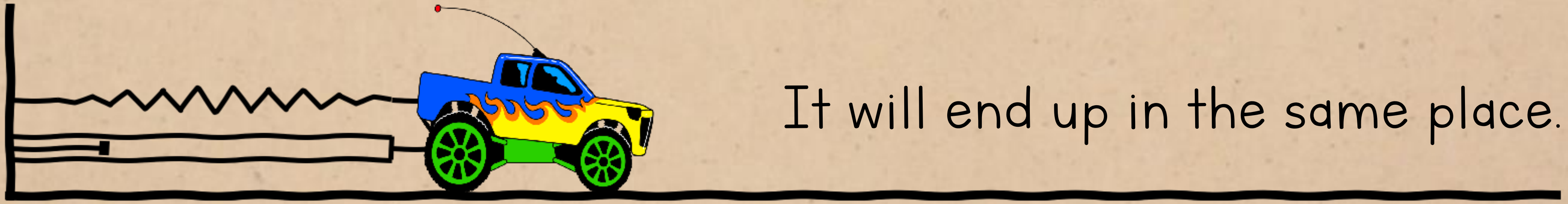
So the initial values affect the transient response, but not the steady-state.

$x(0) = 2.0,\ \dot{x}(0) = 2.0$

FREE RESPONSE

No matter where the system starts off, or how fast it's going…





It will end up in the same place.